

As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.

Learn more: [PMC Disclaimer](#) | [PMC Copyright Notice](#)

Author Manuscript

Peer reviewed and accepted for publication by a journal



Biochim Biophys Acta. Author manuscript; available in PMC: 2007 Oct 16.

Published in final edited form as: *Biochim Biophys Acta*. 2007 May 22;1773(8):1311–1340. doi: [10.1016/j.bbamcr.2007.05.003](https://doi.org/10.1016/j.bbamcr.2007.05.003)

Function and Regulation in MAPK Signaling Pathways

Lessons Learned from the Yeast *Saccharomyces cerevisiae*

[Raymond E Chen](#)¹, [Jeremy Thorner](#)^{1,*}

[Author information](#) [Article notes](#) [Copyright and License information](#)

PMCID: PMC2031910 NIHMSID: NIHMS28620 PMID: [17604854](#)

The publisher's version of this article is available at [Biochim Biophys Acta](#)

Abstract

Signaling pathways that activate different mitogen-activated protein kinases (MAPKs) elicit many of the responses that are evoked in cells by changes in certain environmental conditions and upon exposure to a variety of hormonal and other stimuli. These pathways were first elucidated in the unicellular eukaryote *Saccharomyces cerevisiae* (budding yeast). Studies of MAPK pathways in this organism continue to be especially informative in revealing the molecular mechanisms by which MAPK cascades operate, propagate signals, modulate cellular processes, and are controlled by regulatory factors both internal to and external to the pathways. Here we highlight recent

advances and new insights about MAPK-based signaling that have been made through studies in yeast, which provide lessons directly applicable to, and that enhance our understanding of, MAPK-mediated signaling in mammalian cells.

1. Introduction

A fundamental property of living cells is the ability to sense and respond appropriately to changing environmental conditions and various other stimuli. One frequently utilized molecular device for eliciting these responses is the three-tiered cascade of protein kinases known as the mitogen-activated protein kinase (MAPK) module [1]. Our current understanding of MAPK pathways is based in large part on research that was conducted first in the eukaryotic microbe, *Saccharomyces cerevisiae* (also known as baker's yeast or budding yeast). Many of the components of these pathways and the mechanisms by which they operate were first identified and characterized in this organism and are now known to have been conserved during the evolution of the entire eukaryotic kingdom. This yeast has served its pathfinding role because it is highly amenable to genetic, biochemical, and cell biological studies, and was the first eukaryote to have its entire genome sequenced.

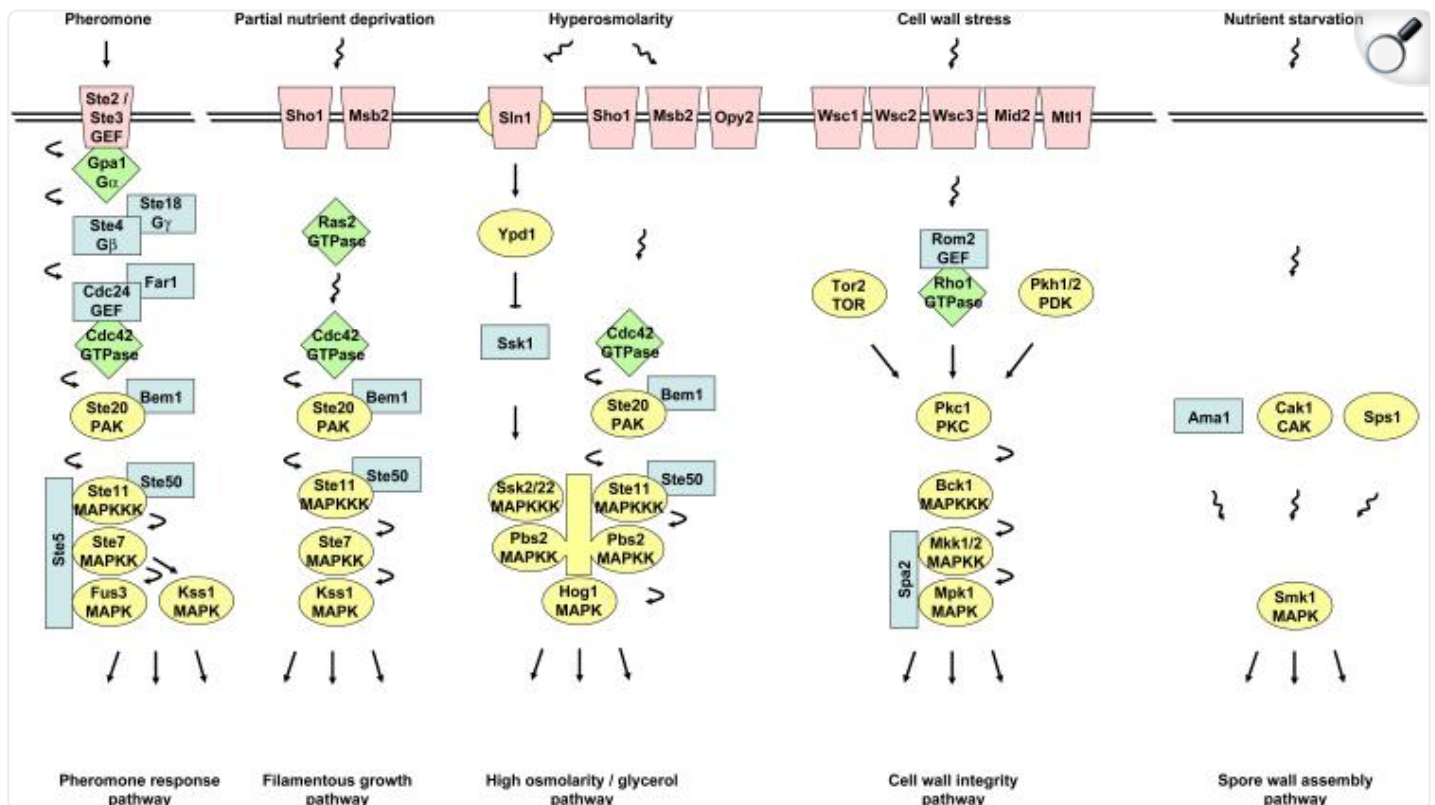
In this article, we begin with an overview of the MAPK pathways in *S. cerevisiae* and the mechanisms of their activation in response to signals or stresses. We then discuss the mechanisms by which these pathways regulate downstream molecular and cellular processes. We also consider the mechanisms by which these pathways are themselves regulated by components both internal and external to the core signal transduction machinery of each pathway. Throughout, our emphasis is on new insights that have been gleaned from studies during the last few years and, importantly, on how the molecular mechanisms and general principles unveiled by these recent studies continue to illuminate previously unappreciated features of MAPK signaling that are more difficult to discern in more complex organisms.

2. Core activation modules

2.1 MAPK cascades

The canonical MAPK pathway contains a key, three-component signal relay in which an activated MAPK kinase kinase (MAPKKK or MEKK) activates a MAPK kinase (MAPKK or MEK), which then

activates a MAPK (or ERK, for extracellular signal-regulated kinase) ([Fig. 1](#)). MAPKKKs contain an N-terminal regulatory domain and a C-terminal serine/threonine protein kinase domain. Upon activation, a MAPKKK phosphorylates two serine or threonine residues at conserved positions in the activation loop of its target MAPKK, which is a dual-specificity (serine/threonine and tyrosine) protein kinase. The activated MAPKK then proceeds to phosphorylate both the threonine and tyrosine residues of a conserved -Thr-X-Tyr- motif in the activation loop of its target MAPK. These phosphorylations activate the MAPK by causing substantial conformational changes, and point mutations in which these phosphoacceptor residues are changed to acidic residues (Glu or Asp) do not suffice to activate MAPKs. In contrast, such phosphomimetic mutations are frequently able to confer constitutive activity to other sub-classes of protein kinases, including MAPKKs. MAPKs are serine/threonine protein kinases in the same CMGC group as cyclin-dependent kinases (CDKs) and phosphorylate their substrates at -Ser/Thr-Pro- motifs.

Figure 1. Schematic diagrams of the MAPK signaling pathways in *S. cerevisiae*.
[Open in a new tab](#)

Symbols are: protein kinases, ovals; GTP-binding proteins, diamonds; scaffold, adaptor, and activating proteins, rectangles; cell surface proteins, trapezoids; activation, arrows; inhibition, T-bars; direct action, smooth lines; indirect action (or unknown molecular mechanism), squiggly lines. For clarity, not all factors and interactions are shown, connections to other pathways and processes upstream of the MAPKs are omitted, and direct targets of the MAPKs are not included (see the text for these details).

To initiate a MAPK cascade, the MAPKKK must be activated. Upstream events that can lead to MAPKKK activation include processes such as occupancy of receptors coupled to heterotrimeric G proteins by their cognate agonists and the binding of the appropriate ligands to other classes of receptors that stimulate production of activated monomeric G-proteins, or both. In contrast to MAPKKKs and MAPKKs, for which a paucity of physiologically relevant substrates have been described (other than their MAPKK and MAPK targets, respectively), MAPKs phosphorylate a

diverse set of well-characterized substrates, including transcription factors, translational regulators, MAPK-activated protein kinases (MAPKAP kinases), phosphatases, and other classes of proteins, thereby regulating metabolism, cellular morphology, cell cycle progression, and gene expression in response to a variety of extracellular stresses and molecular signals.

2.2 The Cdc42-PAK module: G-proteins, protein kinases, and adaptors

Three of the MAPK pathways present in yeast are activated by a common agent, namely, a member of the p21-activated protein kinase (PAK) family of protein kinases, Ste20 ([Fig. 1](#)). In this case, the p21 is the small, monomeric Ras-related GTPase, Cdc42. Ste20 is activated by Cdc42 as follows: the C-terminal kinase domain of Ste20 is held in an inactive state by association with an autoinhibitory sequence present in its N-terminal domain that overlaps with a Cdc42/Rac interactive binding (CRIB) motif; binding of active (GTP-bound) Cdc42 to the CRIB motif relieves this autoinhibition ([2](#)). In all three pathways, activated Ste20 is responsible for phosphorylating and activating Ste11, and thus serves as a MAPKKK kinase (MAPKKKK) ([3](#), [4](#)). Hence, in each pathway, upstream events must stimulate GTP loading of Cdc42 and place Ste20 in close proximity to GTP-bound Cdc42. Much of the Cdc42 in the cell is permanently located at the plasma membrane due to geranylgeranylation of its C-terminal -CAAX box. Ste20 is brought to the same general vicinity, in part, via its binding to Bem1, an adaptor protein that interacts with proline-rich motifs in Ste20 through its tandem N-terminal Src-homology-3 (SH3) domains and is also membrane-tethered via an internal phosphoinositide-binding Phoxhomology (PX) domain ([5](#)). Importantly, however, in each pathway, membrane recruitment of Ste20 and its juxtaposition to its substrate, Ste11, is also facilitated by upstream pathway-specific components (see Section 3). For example, the C-terminal tail of Ste20 has a high-affinity binding site for the G $\beta\gamma$ complex (Ste4-Ste18) released in the response pathway that is triggered by the binding of G-protein-coupled receptors (GPCRs) to peptide mating pheromones ([6](#)). G $\beta\gamma$ is firmly anchored to the plasma membrane via both S-palmitoylation and S-farnesylation of the C-terminal -CCAAX box of G γ (Ste18). Thus, Ste20-G $\beta\gamma$ interaction ensures that this MAPKKKK will be recruited most efficiently to the region of the plasma membrane containing the highest number of ligand-occupied pheromone receptors.

Likewise, Ste11 is recruited to the same general vicinity as activated Ste20 because it interacts with a small adaptor protein, Ste50. Ste11 associates with Ste50 via heterotypic interaction of an N-terminal sterile-alpha-motif (SAM) domain with a SAM domain at the N-terminus of Ste50 ([7](#)). Ste50, in turn, is able to associate via its C-terminal Ras-association (RA) domain with Cdc42 ([8](#),

[2\]](#), thereby tethering Ste11 at the plasma membrane. However, membrane recruitment of Ste11 and its propinquity to Ste20 are imposed by additional pathway-specific factors. For example, in the pheromone response pathway, a scaffold protein, Ste5, binds both Ste11 [[10-12](#)] and the membrane-tethered G $\beta\gamma$ complex [[13](#)], thereby delivering Ste11 to the membrane site with the greatest number of occupied pheromone receptors. As another example, in the High-Osmolarity-Glycerol (HOG) pathway required for survival in response to hyperosmotic stress ([Fig. 1](#)), both Ste11 and Ste50 are able to bind the cytosolic tail of a polytopic transmembrane protein, Sho1, that is one component of the primary osmosensor [[9](#), [14](#)].

3. MAPK signal transduction pathways

The genome of *S. cerevisiae* encodes multiple MAPKs [[15](#)] that possess the diagnostic -T-x-Y- in the activation loop and other hallmark features of this class of enzyme ([Fig. 1](#)). One (Fus3; -TEY-) mediates cellular response to peptide pheromones. Another (Kss1; -TEY-) permits adjustment to nutrient limiting conditions. A third (Hog1; -TGY-) is necessary for survival under hyperosmotic conditions. A fourth (Slt2/Mpk1; -TEY-) is required for repair of injuries to the cell wall. Another, still poorly characterized, but clearly MAPK-like and Slt2/Mpk1-related, kinase (Ykl161c; -KGY-) is also thought to contribute to the processes that maintain cell wall integrity. A fifth (Smk1; -TNY-), along with another, more divergent MAPK-related kinase (Ime2; -TAY-), regulates spore wall assembly during meiosis and sporulation, a developmental response of *MATa*/*MATa* diploid cells to acute nutrient deprivation. Below, our current picture of each of these MAPK pathways is described. In those cases where a MAPK pathway acts in conjunction with other independent signaling pathways to yield a composite response to a given stimulus, our treatment focuses on the MAPK branch and its contribution to the output. Detailed discussion of the downstream effectors of the activated MAPKs are deferred to Section 4, where we consider mainly those substrates whose functions are, relatively speaking, the best understood at the mechanistic level.

3.1 Pheromone response pathway

S. cerevisiae exists in two haploid cell types, *MATa* (a cell, for short) and *MAT α* (α cell, for short). Like the gametes of multicellular organisms, an a cell and an α cell can mate by undergoing cellular and nuclear fusion to generate a third cell type, the *MATa*/*MAT α* diploid (a/ α cell, for short). Mating is the end result of a complex series of changes in cellular physiology that are all initiated in response to peptide pheromones secreted by the haploid cells. The a cells release a-factor, a C-terminally farnesylated 12-residue peptide that acts on the α cells; the α cells release α -

factor, an unmodified 13-residue peptide that acts on the **a** cells. The **a**-factor acts on **a** cells by binding to the GPCR Ste2; and, **a**-factor acts on **α** cells by binding to the GPCR Ste3. Both pheromone receptors are coupled to a common heterotrimeric G protein, Gpa1-Ste4-Ste18, where Gpa1 is Ga and Ste4-Ste18 is the Gβγ complex, as mentioned earlier. As recounted below, events initiated by engagement of these GPCRs by their cognate pheromones leads to activation of Cdc42 and, eventually, to activation of the MAPK, Fus3. The action of Fus3 is responsible for eliciting the expression of numerous mating-specific genes, imposing cell cycle arrest, promoting polarized cell growth to form copulatory projections toward the mating partner (cells that have undergone this morphological transition are called “shmoo”), establishing the changes in the plasma membrane and cell wall necessary for cell-cell fusion (plasmogamy), and orienting the nucleus and modifying its envelope to permit fusion of the two haploid nuclei (karyogamy). Both the heterotrimeric G protein and Cdc42 also act through additional effectors to stimulate other branches of the response machinery that are necessary to produce mating-competent cells and achieve optimally efficient mating [16-21]. Thus, yeast pheromone response is clearly a network of interlocking events, rather than a simple linear pathway, and is arguably one of the best understood MAPK-based signal-response systems in biology. For other recent reviews, see [22-26].

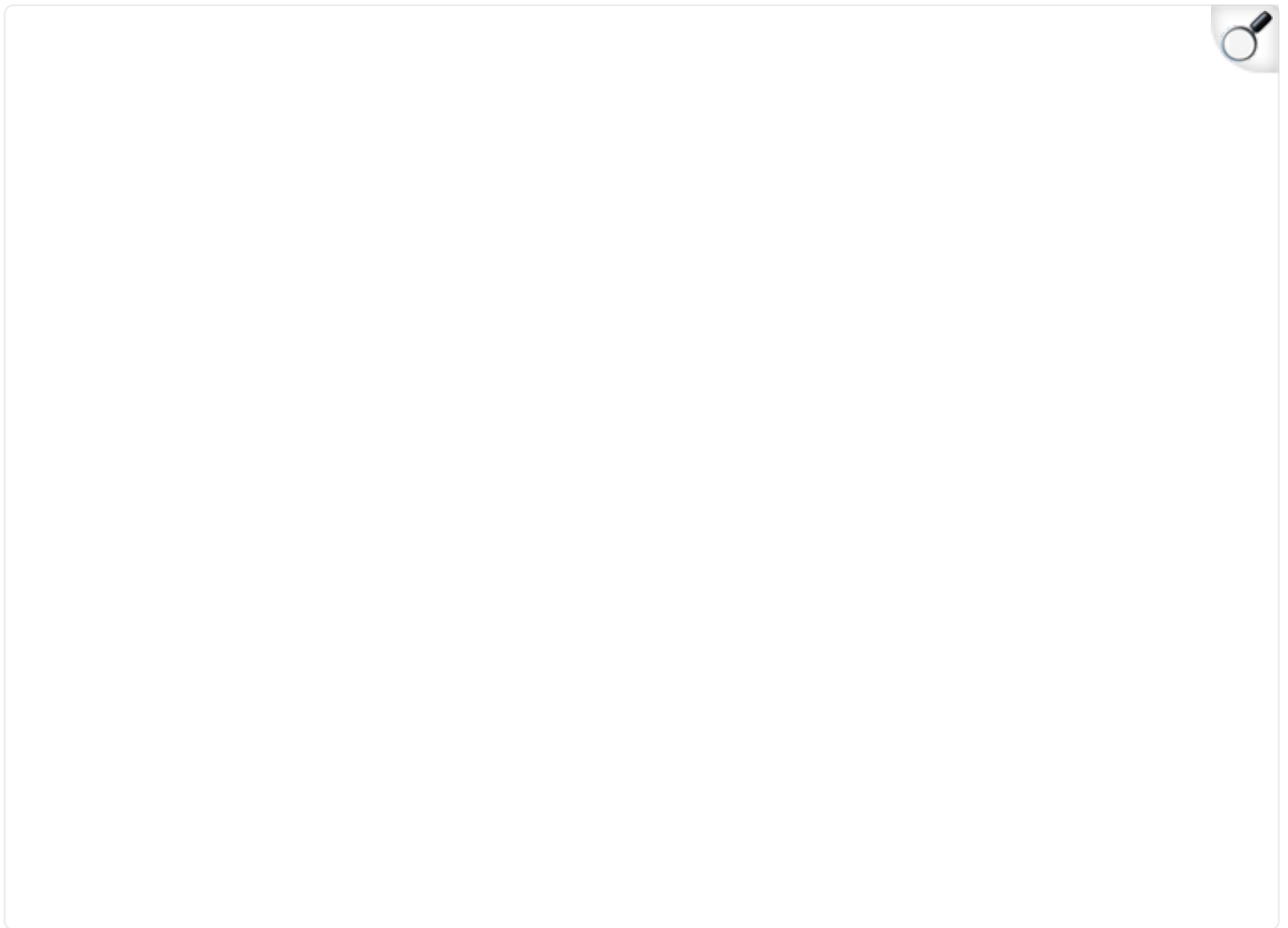
The pheromones and pheromone receptors are the only cell-type specific components in the mating pathway. In the canonical manner, binding of a pheromone to its cognate GPCR allows the receptor to serve as a guanine nucleotide exchange factor (GEF) on its coupled heterotrimeric G-protein, facilitating the release of GDP and the subsequent binding of GTP by Gpa1 (Ga subunit). GTP binding to Ga alters its interaction with Gβ (Ste4), dissociating Gpa1 from the Gβγ complex [27]. Like the Gβγ, Gpa1 remains tethered to the plasma membrane by lipophilic modifications, in this case, N-myristoylation and S-palmitoylation. The newly exposed surfaces of the released Gβγ can now interact with three known effectors: Ste20 and Ste5, as mentioned earlier, and a protein weakly related to Ste5, Far1. Like Ste5 [28, 29], Far1 associates with Gβγ via its N-terminal RING-H2 domain [19]. However, unlike Ste5, the C-terminus of Far1 binds to, and most likely activates, Cdc24, which is the only known GEF for Cdc42 [19, 20, 30]. Cdc24, in turn, has an inherent propensity, first, to associate with the plasma membrane because it contains an internal phosphoinositide-binding pleckstrin-homology (PH) domain and, second, to localize near Ste20 because its C-terminus contains a Phox-and-Bem1-binding (PB1) domain [31]. As mentioned earlier, Ste20 is the Bem1-binding PAK that is activated by Cdc42-GTP and serves as the MAPKKKK to phosphorylate and thereby trigger activation of the MAPKKK, Ste11, initiating activation of the remainder of the MAPK cascade, namely Ste7 (MAPKK) and Fus3 (MAPK) (Fig. 1). Ste5 is a scaffold protein that binds all three component kinases of the cascade (Ste11, Ste7, and Fus3) [10-12]. In addition to the affinity of its N-terminal RING-H2 domain for Gβγ, membrane

recruitment of Ste5 also requires a short N-terminal amphipathic α -helix (PM motif) [32] and, like Far1 and Cdc24, an internal PH domain [33]. In any event, because both Cdc42 and G β γ are firmly plasma membrane-anchored [34-36], the ability of free G β γ to bind its three effectors—Far1, Ste20, and Ste5—promotes encounter of Cdc42 with its activator (Cdc24) and places its target kinase (Ste20) and the downstream cascade that needs to be activated (Ste11, Ste7 and Fus3) in close juxtaposition and at high local concentration.

In addition to Fus3, pheromone stimulation also leads to transient activation of another MAPK, Kss1 [37, 38]. Activation of Kss1 also occurs via Ste11 and Ste7, but is not dependent on the scaffold protein, Ste5 [39-41] (see also Section 5.4.1). Based on their primary structures, Fus3 and Kss1 are the most closely related pair of *bona fide* MAPKs in the yeast genome and appear to be the orthologs of mammalian Erk1 and Erk2, respectively. Cells lacking both Fus3 and Kss1 are sterile, whereas the presence of either one alone permits mating, indicating that these MAPKs have a redundant function. However, this overlap in function is only partial because quantitative analysis shows that loss of Kss1 does not measurably reduce mating proficiency, whereas loss of Fus3 reduces mating efficiency to ~10% of the wild-type level [37, 42]. Analysis of other indicators (cell cycle arrest, morphological changes, gene induction patterns) of signal throughput in cells lacking either Fus3 or Kss1 [42-44] indicate that Fus3 is responsible for the majority, but not the entirety, of the MAPK-dependent pheromone response. In contrast, Kss1, but not Fus3, is essential for the invasive growth response in haploids and the pseudohyphal growth response in diploids, which we will refer to here, for consistency and simplicity, as the filamentous growth response [45] (see Section 3.2). This situation raises important questions about the exact nature of the relative contributions of Fus3 and Kss1 during normal pheromone response and about how cells selectively signal through Kss1 during the filamentous growth response, issues that we return to and discuss further in Section 5.4.

In this context, however, it should be appreciated that Fus3 is much more efficient than Kss1 at mediating pheromone-induced cell cycle arrest [42], most likely because Fus3 phosphorylates Far1 more efficiently [46, 47] due to a high-affinity docking site in Far1 that binds Fus3, but not Kss1 [48]. Phosphorylated Far1 functions as an inhibitor of G1 cyclin-bound CDK (Cln-Cdc28) in a manner distinct from its role in MAPK activation (see Section 4.3). Fus3 also serves as a negative regulator of filamentous growth because, unlike Kss1, it phosphorylates and leads to the degradation of the Tec1 transcription factor necessary for induction of the genes involved in this developmental outcome (see Section 5.4.2. and Fig. 4).

Figure 4. Mechanism of MAPK-directed developmental commitment and cell fate determination in yeast.



[Open in a new tab](#)

Fus3 (in response to pheromone) specifically phosphorylates and promotes the ubiquitin-dependent and proteasome-mediated destruction of the transcription factor, Tec1, which is uniquely required for induction of the genes necessary for filamentous growth (see [Fig. 3](#)). The same site in Tec1 is not a substrate for Kss1. In this way, even though Kss1 is also activated (albeit transiently) in response to pheromone, a filamentation response is precluded. Hence, this process provides an elegantly simple mechanism to ensure that, once a haploid cell is exposed to pheromone, it is irrevocably committed to the mating response. Similar mechanisms are likely involved in directing the development transitions and in dictating the cell fate commitments observed in mammalian cell differentiation.

These examples suggest that the qualitatively different contributions of Fus3 and Kss1 to the events required for mating may be due to differences in substrate preference or differences in the temporal and spatial dynamics of the two MAPKs themselves, another issue to which we will return. Of potential interest in the latter regard is some evidence suggesting that signal transduction in response to pheromone is noisier (shows greater variance in output across individual cells of a population) when mediated by Kss1 than when mediated by Fus3 [49]. If this property reflects an intrinsic difference between how Kss1 and Fus3 propagate a signal, it suggests that a population of cells may be able to explore a wider range of functional states during nutrient limitation, which may be of physiological importance for the ability of the population to survive, than during exposure to a pheromone stimulus. Additionally, because *FUS3* is a pheromone-inducible gene (but *KSS1* is not) the ratio of active Fus3-to-Kss1 increases with increasing pheromone concentration and time after exposure to pheromone [43, 49], raising the possibility that early on, or when exposed to a low or spurious level of pheromone, cells may be able to initiate some responses before they commit to the growth arrest and other processes required for mating [50].

3.2 Filamentous growth pathway

In environments containing ample nutrients, *S. cerevisiae* cells are ovoid and proliferate by budding. Under these conditions, a haploid mother cell always buds off new daughter cells from the same cell pole as, and adjacent to, its own birth end (a pattern referred to as axial budding); a diploid mother cell buds off new daughters from either its birth end or the opposite cell pole, alternating about 50% of the time (a pattern referred to as bipolar budding). In environments where nutrients have become limiting, the cells undergo morphological changes and become more elongated and proliferate in a unipolar pattern, in which new daughter cells arise only at the cell pole opposite the birth end of their mother. Additionally, cells growing in such conditions exhibit increased cell-cell adhesion, increased cell-substratum adhesion, and an increased ability to penetrate their substratum. It is thought that the combination of cell elongation, cell-cell and cell-surface adhesion, substratum invasion, and highly directional growth, i.e. division away from (as opposed to next to) existing cells, serves to permit a colony of yeast cells, wherein each individual cell is non-motile, to spread out as a means to better explore its surroundings for additional nutrients. By contrast, if nutrient supply is not limiting, the combination of a more round cell shape, decreased adhesiveness, and division near existing cells, presumably facilitates more rapid occupancy of the niche by the population.

As mentioned earlier, these sets of nutrient limitation-induced behaviors are termed pseudohyphal growth in diploids and invasive growth in haploids. Although there are important biological and mechanistic differences between the two—for example, the former is elicited when nitrogen becomes limiting, whereas the latter is evoked when glucose (the preferred carbon source for *S. cerevisiae*) becomes limiting—many of the primary molecular components and regulatory pathways involved in these filamentous growth responses are the same [45]. Optimum filamentous growth requires the action of at least three distinct classes of protein kinases: a 5'-AMP-dependent protein kinase (AMPK), Snf1; a specific isoform of 3', 5'-cyclic AMP-dependent protein kinase (PKA), Tpk2; and, as already introduced, the MAPK Kss1. Similar to Fus3, the other two PKA isoforms encoded in the *S. cerevisiae* genome, Tpk1 and Tpk3, are negative regulators of filamentous growth. Recent reviews of filamentous growth and its regulation can be found in [38, 45, 51-56].

Activation of Kss1 requires Ste20 (PAK), Ste11 (MAPKKK), and Ste7 (MAPKK) [9, 37] (Fig. 1). Because Ste20 activation requires GTP-bound Cdc42, there must be some mechanism to bring its GEF, Cdc24, to the plasma membrane (and, perhaps, activate it) during filamentous growth. In pheromone response, this membrane delivery, localization and activation of Cdc24 is accomplished, in large part, by its association with Far1, the scaffold protein that is recruited to the plasma membrane via its binding to free G $\beta\gamma$. Activation of Cdc42 during filamentous growth is known to be dependent on active Ras2 [57], yeast homolog of mammalian H-Ras. However, how Ras2 action promotes membrane recruitment and activation of Cdc24 to stimulate GTP loading on Cdc42 is not clear. Furthermore, production of activated Ras2 presumably demands that something about the conditions that promote filamentous growth also stimulates membrane recruitment and activity of the GEF for Ras2, Cdc25. The precise mechanisms by which Cdc25 and Ras2 become activated in this pathway are also unclear. However, several distinct transmembrane proteins that reside in the plasma membrane and are exposed to the cell surface are necessary (in some cases, in haploids, and in other cases, in diploids) for initiation of filamentous growth. These transmembrane proteins include: Sho1 [58] (four transmembrane segments); Msb2 [58] (one transmembrane segment); Mep2 [59, 60] (ten transmembrane segments); and, Gpr1 [61, 62] (seven transmembrane segments).

Sho1 can form hetero-oligomeric complexes with Msb2, and the absence of either protein blocks Kss1 activation and prevents filamentous growth in haploids [58]. Interestingly, Sho1 can also form hetero-oligomeric complexes with another single-pass transmembrane protein, Opy2, and absence of either protein blocks activation of the Hog1 MAPK and the HOG response [63, 64], which is necessary for continued growth under hyperosmotic conditions (see Section 3.3).

However, loss of Msb2 does not prevent HOG response [65], and loss of Opy2 does not block filamentous growth (E.S. Klimenko and J. Thorner, unpublished results). Thus, Sho1 serves as a common subunit of two different membrane sensors that allow cells to respond to two different stimuli. The role of Sho1 is reminiscent of what is seen for several classes of cell surface receptors in animal cells, such as the common gamma chain (γ c) shared by different multi-chain cytokine receptors [66].

Msb2 possesses a large highly O-glycosylated exocellular domain that is related to the so-called mucin family of mammalian transmembrane proteins. Strikingly, deletions within the extracellular mucin-homology domain of Msb2 cause significant constitutive activation of the filamentous growth response in haploids [58]. This observation leads to the simple model that glucose limitation leads to under-glycosylation of Msb2, alleviating some negative structural constraint and promoting the events necessary to trigger downstream signal propagation. In this regard, it has also been reported that the short C-terminal cytosolic tail of Msb2 can bind to Cdc42 directly [58], but the data in support of this particular claim are unconvincing.

Mep2 is a high-affinity ammonia permease [67] that also acts as a nitrogen sensor and is required for diploid pseudohyphal growth. Loss of the related, but lower affinity, ammonia permeases, Mep1 and Mep3, has no effect on diploid filamentation. Activated Ras2 bypasses the need for Mep2 in diploid pseudohyphal growth [59, 60], suggesting that this is the level at which the function of Mep2 is connected to stimulation of the MAPK cascade that activates Kss1 (and PKA; see next paragraph).

Gpr1 is a glucose (and sucrose)-binding GPCR [68] that associates with a distinct G α subunit, Gpa2 [69], and is thought to serve as a carbon sensor [62]. Intriguingly, expression of the *GPR1* gene is also induced under conditions of nitrogen limitation; thus, under limiting nitrogen, the cell presumably becomes more acutely “aware” of the status of its carbon supply. In any event, it was initially thought that Gpa2 associates with Gpr1 in a heterotrimer with either Gpb1/Krh1 or Gpb2/Krh2, two alternative, non-canonical Kelch-repeat-containing G β mimics (as opposed to a classical G β comprising a seven-bladed WD-40 repeat propeller) and a non-canonical non-prenylated G γ -like protein, Gpg1 [70, 71]. There is consensus from several labs that Gpr1 and Gpa2 act upstream of and are necessary for optimal PKA function; however, there is some dispute with regard to the level at which Gpa2 acts. Some evidence suggested that Gpa2 somehow promotes Ras2 activation and Ras2-GTP, a known activator of adenylate cyclase in yeast [72], stimulates cAMP production, which activates PKA by dissociating the inhibitory cAMP-binding regulatory (R) subunit (Bcy1 in *S. cerevisiae*) from the catalytic (C) subunits (Tpk1, Tpk2 and

Tpk3 in *S. cerevisiae*). Other evidence suggested, however, that Gpa2 stimulated PKA independently of any effect on Ras2 or adenylate cyclase *per se* [69]. Recent work has done little to resolve this controversy. At least one group reports that Gpb1 and Gpb2 bind to and stabilize the Ras GTPase-activating proteins (GAPs), Ira1 and Ira2, and thus that Gpa2-dependent sequestration of Gpb1 and Gpb2 will cause a reduction in Ira1 and Ira2 that will, in turn, elevate the level of Ras2-GTP, which can then stimulate adenylate cyclase and cAMP production, leading to higher PKA activity [73]. However, at least two other groups [74, 75] present convincing evidence that Gpb1 and Gpb2 are novel subunits of the inactive R subunit-bound state of PKA. Moreover, they show that GTP-Gpa2 has a high affinity for these negative regulators and removes them, leading to a significant increase in basal PKA activity and to a significant reduction in the level of cAMP necessary to fully activate PKA. The mechanisms by which the signals transduced by all of the transmembrane proteins discussed above are coordinated to achieve an optimal filamentous growth response are not known.

In this same regard, it has also been observed that filamentous growth can be stimulated by fusel alcohols [76, 77] and aromatic alcohols [78]. These small molecules are, of course, potential membrane perturbants and may thus act via effects on one or more of the membrane proteins discussed immediately above. Nevertheless, the fact that these small molecules are secondary metabolites generated by the yeast itself and released into the surrounding milieu has led to the proposal that these compounds could provide a quorum-sensing mechanism for regulating the onset of filamentous growth [78].

Finally, Snf1 (AMPK) has a critical role in shifting the transcriptional program of yeast cells to deal with alternative carbon sources when glucose becomes limiting [79]. Not surprisingly, therefore, Snf1 also has a role in promoting both invasive growth in haploids and pseudohyphal growth in diploids in response to glucose depletion [80]. However, this role is not simply an indirect one of establishing the appropriate metabolic conditions to permit continued growth when glucose carbon is scarce. Indeed, genetic evidence indicates that Snf1, in association with a specific one of its three different β -subunit isoforms (Gal83), phosphorylates and antagonizes two repressors, Nrg1 and Nrg2, thereby increasing expression of the *MUC1/FLO11* gene [81], which encodes a GPI-anchored cell surface glycoprotein that is important for the cell-cell and cell-substratum adhesion required for filamentous growth [82]. Recent evidence indicates that control of Snf1 function by yeast TOR may contribute to how nitrogen supply regulates pseudohyphal growth in diploids [83].

3.3 High osmolarity / glycerol pathway

An increase in the dissolved solute concentration of the extracellular medium to a level higher than the internal osmolarity of the cell causes a drop in turgor pressure that may be sufficiently deleterious to threaten cell viability in the absence of a mechanism to restore osmotic balance. To increase the internal osmolyte concentration in a relatively innocuous way as a means to combat external hypertonic stress, yeast cells increase their synthesis of glycerol, a highly water soluble and inert solute. This mechanism is referred to as the High-Osmolarity-Glycerol (HOG) response. Survival under hyperosmotic conditions via the HOG pathway requires activation of the eponymous Hog1, whose functional ortholog in mammalian cells is the p38 family of stress-activated MAPKs (SAPKs) [84]. For other recent reviews of HOG pathway signaling, see [85-89].

Two distinct upstream inputs can lead to activation of Hog1 (Fig. 1). The first route involves a histidine-aspartate phospho-relay module similar to those utilized in bacterial two-component signaling systems. An apparent osmosensor, Sln1, which contains two transmembrane segments and resides in the plasma membrane, also contains a histidine kinase domain within its cytoplasmic C-terminal segment. Under iso-osmotic conditions, Sln1 is active and catalyzes autophosphorylation and subsequent phospho-transfer to an intermediate protein, Ypd1, which transfers the phosphate group to an aspartate residue on a response regulator, Ssk1 [90], preventing interaction of Ssk1 with two semi-redundant MAPKKKs, Ssk2 and Ssk22. Mild hyperosmotic stress inhibits Sln1, resulting in an increase in the amount of unphosphorylated Ssk1. Unphosphorylated Ssk1 is able to bind to and activate Ssk2 and Ssk22 [91]. These MAPKKKs phosphorylate a dedicated MAPKK, Pbs2, which in turn, is responsible for dual phosphorylation and activation of the MAPK, Hog1 [63, 92].

The second route by which Hog1 can be activated does so via the alternative MAPKKK, Ste11, which we also encountered in both the pheromone response pathway and the filamentous growth pathway (Fig. 1). The “tricks” necessary here are to steer active Ste11 toward Pbs2 and prevent it from encountering Ste7. These maneuvers seem to be accomplished by fixing a fraction of the Ste11 in firm association with the plasma membrane via contacts with multiple components of the upstream machinery necessary to trigger a response to severe hyperosmotic stress. First, it has been reported that Ste11 binds directly to the C-terminal cytosolic tail of Sho1 [14]. Second, Ste11 binds tightly to the downstream MAPKK, Pbs2 [93], and Pbs2 itself is bound to Sho1, an interaction mediated by the binding of a proline-rich motif in the N-terminal regulatory domain of Pbs2 to an SH3 at the end of the cytosolic tail of Sho1 [63]. Third, Ste50 is, in essence, a tightly

bound non-catalytic subunit of Ste11 [7, 94], and Ste50 can associate via its RA domain with both a membrane-anchored protein, Cdc42 [8], and an integral membrane protein, Opy2 [64].

The MAPKK Pbs2 represents a true node shared between the Sln1-dependent and the Sho1-dependent branches of the HOG pathway inputs. The N-terminus of Pbs2 contains a high-affinity docking site for the MAPKKKs, Ssk2 and Ssk22, of the Sln1 branch [95] and, as already mentioned above, Pbs2 also associates with Ste11 [93] and with the osmosensor, Sho1 [63]. Moreover, Pbs2 also binds its target MAPK, Hog1, via specific docking motifs distant from the active sites of these two kinases [93, 96, 97]. Thus, it has been suggested [14, 93] that Pbs2 serves the dual function of being the dedicated MAPKK of the HOG pathway and also the platform or scaffold for proper assembly of the signaling complexes necessary to propagate the signals that initiate the HOG pathway in the first place. As expected, stimulation of Ste11 in the Sho1-dependent branch requires the function of Cdc42 and Ste20 [9, 98, 99].

Activation of Hog1 causes its rapid translocation from the cytoplasm to the nucleus [100]. Nuclear Hog1 binds and phosphorylates several transcription factors, interacts with chromatin modifying enzymes and RNA polymerase II, and affects the expression of hundreds of genes in response to hyperosmotic shock [89] (see Section 4.4). Interestingly, however, preventing nuclear localization of Hog1, either by tethering it to the plasma membrane or by deleting the gene for its nuclear import factor, Nmd5, or both, does not render cells osmosensitive [101]. One explanation for this finding might be that factors downstream of Hog1 are the critical agents that must undergo nucleocytoplasmic shuttling, for example, the MAPKAP kinase, Rck2, a known substrate of Hog1 [102, 103], or the transcription factors that are modified by Hog1, such as Smp1 [104]. However, in these cells, the expression profile for the genes normally induced by hyperosmotic shock closely resembles that of *hog1Δ* cells and not that of wild-type cells [101], raising the intriguing possibility that regulation of transcription may not be the essential function of Hog1 required for osmoresistance. In marked contrast, tethering Fus3 to the plasma membrane via the same means totally blocks the ability of Fus3 to complement the mating defect of a *fus3Δ kss1Δ* double mutant [101].

3.4 Cell wall integrity pathway

The MAPK Slt2/Mpk1 becomes activated under a number of different conditions that stress the structure and function of the yeast cell wall, including hypotonic medium, treatment of cells with glucanases (e.g. Zymolyase), exposure to chitin-binding agents (e.g. Calcofluor White and Congo

Red), as well as oxidative stress, depolarization of the actin cytoskeleton, and pheromone-induced morphogenesis [105, 106]. It is thought that the common element sensed in all of these cases is stretching of the plasma membrane and/or alterations of its connections to the cell wall. The genes under control of this response pathway include many involved in the synthesis and modification of the major components of the yeast cell wall (glucan, mannan, and chitin) [107, 108], and lack of an Slt2/Mpk1-dependent response causes cell lysis in the absence of an osmotic support in the medium [109]. Hence, the Slt2/Mpk1-dependent response is referred to as the cell wall integrity (CWI) pathway [106].

Five plasma membrane proteins (each containing a single transmembrane segment), Wsc1, Wsc2, Wsc3, Mid2, and Mtl1, have been identified as important for activation of the CWI pathway, although the precise mechanisms by which they sense their direct signals/stressors are unclear. The cytoplasmic C-terminal domains of Wsc1 and Mid2 interact with Rom2 [110], one of three GEFs encoded in the *S. cerevisiae* genome (Rom1, Rom2 and Tus1) thought to be specific for the small Ras-homologous GTPase, Rho1 [111]. Like Cdc42, Rho1 is tethered to the plasma membrane by its C-terminal geranylgeranylated -CAAX box and a preceding tract of basic residues that presumably interacts with the phosphates in the head groups of membrane phospholipids (-KKKKKCVLL in Rho1 and -KSKKCAIL in Cdc42) [112, 113]. Cell cycle-specific control of Tus1 via its phosphorylation by two protein kinases, Cdc28/Cdk1 [114] and Cdc5 (ortholog of mammalian Polo kinase) [115], and the resulting local activation of Rho1 is important for the events necessary for actin contractile ring assembly for cytokinesis. Although both Tus1 and Rom2 (and Rom1) possess the Dbl homology (DH)-PH domain organization found in other GEFs for Rho family G-proteins and both contain C-terminal citron homology (CNH) domains, the N-termini of Rom2 (and Rom1) are quite divergent from that of Tus1 and both contain a Disheveled-EGL10-pleckstrin (DEP) domain that Tus1 lacks [116].

The PH domain of Rom2 binds phosphatidylinositol-4,5-bisphosphate (PtdIns4,5P₂) and is required for its stable plasma membrane localization [117, 118] and its DEP domain seems to be required for its association with at least Wsc1 [119]. It is unclear whether interaction of Rom2 with Wsc1 and the other integral membrane proteins that serve as cell wall sensors is solely to further facilitate its recruitment near membrane-anchored Rho1 or has additional activating functions. It is thought that Rom2 shares its essential functions and its role in CWI signaling with the highly related Rho1 GEF, Rom1 (*rom1*Δ and *rom2*Δ single mutants are viable, whereas a *rom1*Δ *rom2*Δ double mutant is inviable); however, loss of Tus1 exacerbates the phenotype of a *rom2*Δ mutant and, conversely, the phenotypes of a *tus1* mutant can be suppressed by Rom2 overexpression, suggesting that Tus1 may also contribute to CWI signaling [120]. Interestingly, in

this regard, *TUS2* serves as a multicopy suppressor of certain *tor2^{ts}* alleles and of a double mutant lacking two kinases (Ypk1 and Ypk2/Ykr2) in a pathway that responds to sphingolipids and acts in parallel to CWI signaling. The Ypk1- and Ypk2-dependent pathway is thought to couple sphingolipid biosynthesis to the CWI pathway as a means to coordinate plasma membrane synthesis with cell wall expansion [121]. However, overexpression of *Tus1* undoubtedly elevates Rho1 activation and, as discussed further below, one of the effectors of Rho1-GTP is the protein kinase, Pkc1. Pkc1 is an essential activator (MAPKKKK) of the MAPK cascade required for CWI signaling— Bck1 (MAPKKK), Mkk1 and Mkk2 (two semi-redundant MAPKKs), and Slr2/Mpk1 (MAPK) (Fig. 1); the MAPKKs and MAPK in this pathway are bound by the scaffold protein Spa2. Elevation of Pkc1-dependent signaling is known to be sufficient to bypass the need for robust Ypk1- and Ypk2-dependent signaling [121].

Nonetheless, there is involved here some complicated nexus between phosphoinositide generation, sphingolipid biosynthesis, Ca^{2+} signaling, and function of the Tor2 kinase (which is thought to phosphorylate a specific site, the so-called C-terminal hydrophobic motif, in Pkc1, Ypk1 and Ypk2/Ykr2, and perhaps other members of the class of AGC kinases and thereby contribute to their activation [122]). The Tor2-containing complex, TORC2 (which also contains Avo1, Avo2, Bit61, Lst8 and Tsc11/RICTOR) [123], contains two additional, PtdIns4,5P₂-binding, PH domain-containing subunits, Slm1 and Slm2, that are essential for viability and necessary (via interaction with Avo2 and Bit62) for anchoring TORC2 to the plasma membrane [124, 125]. TORC2 is involved in regulating actin cytoskeleton polarization and other actin-based processes (e.g. actin-driven endocytosis of nutrient transporters) and, as mentioned above, perturbation of the actin cytoskeleton stimulates CWI signaling [126-128]. The ability of Slm1 and Slm2 to anchor TORC2 at the plasma membrane depends, not surprisingly, on the PH domains in these proteins and the ability of the cell to generate the plasma membrane pool of PtdIns4,5P₂ [124, 125]. Slm1 and Slm2 are themselves substrates for Tor2-dependent phosphorylation and a *slm1 slm2* double mutant exhibits depolarization of the actin cytoskeleton and eventual cell lysis, as expected if TORC2 is unable to function to maintain proper actin architecture and unable to contribute to full activation of Pkc1 and/or Ypk1 and Ypk2/Ykr2 (even basal activity of Pkc1, Ypk1 and Ypk2/Ykr2 requires phosphorylation on their activation loops by the sphingolipid-dependent protein kinases, Pkh1 and Pkh2 [129]). Conversely, overexpression of Slm1 or Slm2 is able to rescue TORC2 mutants lacking the Tsc11 subunit, indicating that tethering TORC2 to the plasma membrane more efficiently promotes its function [127]. Moreover, Slm1 and Slm2 are hyperphosphorylated in response to heat stress (another condition that stimulates CWI signaling), which appears to activate their function, presumably the recruitment of TORC2 [130, 131]. Dephosphorylation (deactivation, most likely) of phospho-Slm1 and phospho-Slm2 is mediated by the

Ca^{2+} /calmodulin-dependent phosphoprotein phosphatase, calcineurin (also known as PP2B) [130, 131].

Direct downstream effectors of Rho1-GTP include a transcription factor of the two-component signaling response regulator-family (Skn7), two β -1,3-glucan synthases (Fks1 and Gsc2), two formins involved in nucleating actin filament formation (Bni1 and Bnr1), a subunit of the secretory vesicle-associated exocyst complex (Sec3), and at least one protein kinase (Pkc1) [106]. It has been proposed that yeast Pkc1 represents an ancestral progenitor of the PKC family now found in mammals [132]. In fact, the C-terminal kinase domain of Pkc1 shares greatest sequence identity with the human Rho-activated protein kinase, PKN2 (formerly PRK2, for PKC-related kinase-2) [133], but its N-terminal regulatory domain does share greatest similarity over its entire length to the so-called “novel” PKC isoforms (nPKCs), PKCdelta, PKCepsilon and PKCtheta, which are phospholipid (usually phosphatidylserine, PtdSer)-dependent and diacylglycerol (DAG)-activated, but not Ca^{2+} -dependent [134]. Indeed, *in vitro*, both Rho1-GTP [135] and DAG plus PtdSer [136] can substantially stimulate Pkc1 activity [136]. Pkc1 initiates the CWI signaling cascade by phosphorylating and activating the MAPKKK, Bck1 [137], which phosphorylates and activates two semi-redundant MAPKKs, Mkk1 and Mkk2 [138], that, in turn, dually-phosphorylate and activate their target MAPK, Slt2/Mpk1 [106].

Slt2/Mpk1 is responsible for stimulating expression of the genes for enzymes and other factors involved in cell wall biosynthesis and remodeling both directly and indirectly [107, 139]. Slt2/Mpk1 stimulates expression of cell wall biosynthesis genes directly via phosphorylation of the transcription factors, Rlm1 [140, 141] and Swi4 [142] (see Section 4.4). Additionally, Slt2/Mpk1 activation is necessary for stimulation of calcium influx through a plasma membrane Ca^{2+} channel (Cch1-Mid1), a response that, in turn, activates calcineurin (a heterotrimeric enzyme comprising two Ca^{2+} -binding regulatory subunits, Cmd1/calmodulin and Cnb1, associated with either of two semi-redundant catalytic subunits, Cna1 and Cna2) [143, 144]. Activated calcineurin dephosphorylates a transcription factor, Crz1 [145], permitting its retention in the nucleus and thereby its ability to stimulate expression of genes involved in dealing both with cell wall stress [146] and with ER stress caused by agents such as the azole drug miconazole (which blocks synthesis of the membrane sterol, ergosterol) and the antibiotic tunicamycin (which prevents glycoprotein biogenesis by blocking synthesis of mannose-rich Asn-linked oligosaccharide chains) [147]. How Slt2/Mpk1 action promotes Cch1-Mid1 channel opening is not clear, but presumably involves phosphorylation of one or the other, or both, of these subunits (and/or of an interacting protein).

With regard to cross-talk between and coordination of distinct MAPK pathways, it has been found recently [148] that Slt2/Mpk1 becomes activated in response to hyperosmotic shock in a manner that depends primarily on the O-glycosylated, integral plasma membrane protein Mid2 (rather than on any of the other CWI sensors) (Fig. 1) and also requires activated Hog1. Mid2 is also required for the activation of Slt2/Mpk1 that is observed when the extracellular medium is rapidly acidified (low pH stress), but the role of Hog1 in this process was not explored [149]. Similar to hyperosmotic stress, perturbation of cell wall β -1,3-glucan by digestion with Zymolyase also activates Slt2/Mpk1 in a Hog1-dependent manner, but requires Sho1 to do so (and none of the “classical” CWI sensors, Wsc1, Wsc2, Wsc3, Wsc4, Mid2, or Mtl1) [150]. At what level activated Hog1 promotes Slt2/Mpk1 activation is not known (C. Bermejo-Herrero, personal communication).

The Srb10/Ssn3/Cdk8-Srb11/Ssn8/cyclin C complex phosphorylates the C-terminal- repeat-domain (CTD) of the largest subunit (Rpo21) of RNA polymerase II and thereby represses transcription of a large number of genes. Curiously, when Slt2/Mpk1 is activated by exposure of the cell to reactive oxygen species (oxidative stress), but not when this MAPK is activated by other means (e.g. heat stress), cyclin C is destroyed in a manner that depends on Slt2/Mpk1 [151]. Cells lacking Slt2/Mpk1 are hypersensitive to the growth inhibitory effects of oxidants, and absence of cyclin C (but not loss of Cdk8) suppresses the oxidative hypersensitivity of *slt2 Δ* (*mpk1 Δ*) cells, suggesting that Slt2/Mpk1-mediated destruction of cyclin C does something other than simply eliminate Cdk8-cyclin C-dependent transcriptional repression [151].

Finally, both Slt2/Mpk1 and Hog1 appear to be among the clients of yeast HSP90 (Hsc82 and Hsp82) and associate with this chaperone only when the kinases are in their active, phosphorylated state [152, 153]. This interaction also seems to require the essential HSP90 co-chaperone, Cdc37 [154, 155]. A point mutation in Hsp90 was identified that permits normal Slt2/Mpk1 activation upon heat shock or caffeine treatment, but abolishes Rlm1-dependent transcription [152]. This mutant Hsp90 also rescues the inviability of strains expressing a hyperactive Mkk1 allele, although it is unclear whether the mutant Hsp90 is no longer able to interact with Slt2/Mpk1.

3.5 Spore wall assembly pathway

Upon deprivation of both a fermentable carbon source and an additional essential nutrient (nitrogen, phosphorus, or sulfur) [156], *MATa/MATa* diploids undergo meiosis and enclose the

resulting haploid nuclei within coats composed of four layers, yielding desiccation-, heat- and solvent-resistant spores. The innermost two layers are composed primarily of glucan and mannan, respectively, and resemble those same layers in the walls of vegetative cells. The outer two layers are spore-specific and are primarily composed, respectively, of chitin (linear β -1,4-linked N-acetyl-D-glucosamine)/chitosan (linear β -1,4-linked D-glucosamine) and proteins cross-linked by dityrosine formation. Upon restoration of nutrients, the haploid spores are able to germinate (for review, see [157, 158]).

Diploids lacking the MAPK Smk1 undergo meiosis, but exhibit defective assembly of the outer two spore wall layers [159, 160]. Interestingly, spore wall assembly defects in a *smk1* hypomorph occur at progressively later times as the *smk1* gene dosage is increased, suggesting that different steps in the process may be regulated by quantitative thresholds of Smk1 activity [161]. Smk1 interacts physically with Gsc2, a 1,3- β -glucan synthase subunit required specifically for synthesis of the glucan layer of the spore coat, and negatively regulates its glucan synthase activity. Deletion of *GSC2* rescues the chitosan layer assembly defect of *smk1/smk1* cells, suggesting that in wild-type cells, which deposit the layers of the spore coat sequentially from innermost to outermost, deposition of the chitosan layer may require Smk1-mediated inhibition of Gsc2 activity to terminate Gsc2-dependent synthesis of the glucan layer [160]. Although Chs3, the major chitin synthase in yeast, is known to be required for proper assembly of the chitosan layer [162], it is unclear whether the processes controlled by Smk1 include Chs3 synthesis, function or localization [163]. Likewise, although Smk1 has been shown to be required for proper expression of late sporulation genes, it is still not resolved whether its role in spore wall assembly can be completely explained by its apparent gene regulatory functions [164].

Smk1 is only expressed during sporulation; this timing is regulated at the transcriptional level by a middle sporulation element (MSE) in its promoter [165]. Smk1 is phosphorylated at the canonical Thr-X-Tyr motif in its activation loop, and these residues are required for its function *in vivo* [166]. Remarkably, however, no upstream activators for Smk1 of the MAPKK or MAPKKK class have yet been identified. Smk1 activation is dependent on Ama1, a meiosis-specific activator of the anaphase promoting complex (APC) [167], suggesting that some inhibitory factor needs to be removed, and also on Cak1 (the CDK-activating kinase), but not on Cdc28, the direct and essential substrate of Cak1 during mitosis [166]. Diploids lacking another sporulation-specific kinase, Sps1 (closest mammalian ortholog in size, sequence and overall match length is Osr1), have a phenotype similar to cells lacking Smk1 [168]. It was proposed, therefore, that Sps1 may serve as an upstream activator of Smk1 [169]. However, this suggestion is not likely to be correct because Sps1 is not expressed any earlier than Smk1 during the sporulation program [170, 171] and

because Sps1 localizes to the spore coat itself and is necessary for the recruitment of wall-synthesizing enzymes there, including Chs3 [172]. Moreover, contrary to the view that either Sps1 or Cak1 function upstream of Smk1, overexpression of Cak1 suppresses the phenotypes of certain conditional *smk1* mutants [166] and localization of Chs3 is not perturbed in *smk1/smk1* cells [172]. Thus, even if Smk1 is a direct target of Sps1 and Cak1, both Sps1 and Cak1 clearly have functions separate from their putative role in activating Smk1.

4. Regulation by MAPKs

4.1 Ion transporters

Hyperosmotic shock causes a very rapid dissociation (≤ 1 min) of transcription factors and the transcription machinery not already engaged in elongation from chromatin, which is reversed within 10–30 min [173]. Reassociation of proteins with chromatin is dependent on Hog1 phosphorylation of the cytoplasmic domain of the Na^+/H^+ antiporter, Nha1, a modification that stimulates its ability to extrude Na^+ from the cell. A model was proposed [173] which posits, first, that the physicochemical force of hyperosmotic shock causes an immediate increase in the concentration of Na^+ throughout the cell, leading to general dissolution of protein-DNA interactions in the nucleus and, second, that Hog1-promoted pumping of Na^+ out of the cell via Nha1 permits protein-DNA reassociation, regenerating the chromatin substrate for longer-term transcriptional responses (which are themselves often Hog1-dependent; see Section 4.4). This model seems at odds, however, with a report indicating that Hog1 is required to decrease Nha1-mediated efflux of K^+ upon hyperosmotic shock [174]. Additional studies will be needed to determine whether Hog1-mediated phosphorylation of Nha1 does indeed affect its ion selectivity such that the modified transporter is able to export Na^+ better and K^+ worse. In a similar way, it has been reported recently that Hog1-catalyzed phosphorylation of the aquaglyceroporin, Fps1, is necessary to block the ability of this channel to mediate the influx of the toxic compound, arsenite, and confer cellular resistance to this noxious agent [175].

4.2 Cytoskeleton and cell morphogenesis

Like the pheromone-induced phosphorylation of the transcription factor Tec1 [176, 177], pheromone-induced phosphorylation of a yeast amphiphysin, Rvs167, *in vivo* is dependent upon Fus3, but not Kss1, and Rvs167 can be phosphorylated *in vitro* by Fus3, but not by Kss1 [178]. Rvs167 is also phosphorylated by Pho85, the yeast ortholog of mammalian CDK5 [179], in

complex with at least two of the ten known Pho85-specific cyclins, Pcl1 and Pcl2 [178, 180]. Phosphorylated Rvs167 exhibits decreased binding affinity for the yeast WASP ortholog (Las17/Bee1), which activates the actin-nucleating Arp2-Arp3 complex [181], and for the synaptojanin (PtdIns4,5P₂ 5-phosphatase) Inp52/Sjl2 [182]. It has been proposed that this release of Las17 from Rvs167 is necessary for Las17 to activate the Arp2-Arp3 complex [178]. However, deletion of Rvs167 exhibits no detectable mating defect [183] and, thus, the significance of Rvs167 being a Fus3 substrate is unclear. Perhaps Fus3-mediated displacement of Rvs167 allows one or more of the other amphiphysin-like proteins in the cell to take its place and contribute to the actin polymerization necessary for the polarized growth and the cell envelope changes necessary for cell fusion [184, 185]. Indeed, cells lacking Rvs161, which contains a membrane curvature-inducing BAR domain [186] (but lacks the C-terminal SH3 domain found in amphiphysin and Rvs167), exhibits impaired cell fusion during the mating process [183], and cells lacking either Rvs161 or Rvs167 display reduced viability upon starvation (hence, “Rvs”) for either carbon or nitrogen, or upon hypertonic stress [187, 188]. Hence, it is possible that MAPK phosphorylation of these amphiphysins may also be important for their functions under these stress conditions.

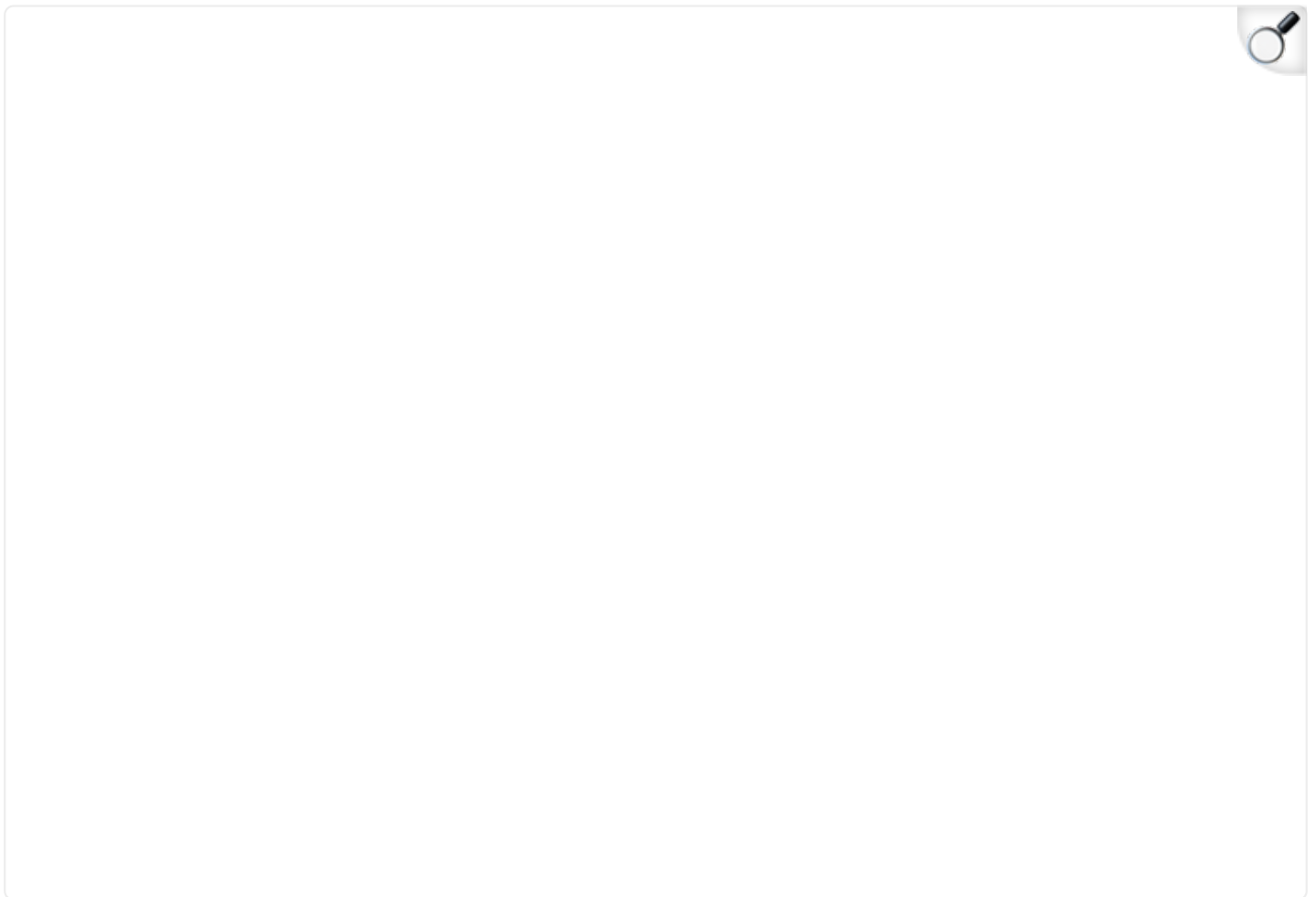
Another direct target of Fus3 is the formin Bni1, which becomes tethered to the tip of the mating projection by interaction with polarisome components [189, 190], interacts with Cdc42-GTP [191], and promotes actin filament assembly in an Arp2-Arp3-independent manner [192]. Fus3 phosphorylates Bni1 *in vitro* and is required for the full phosphorylation and shmoo-tip localization of Bni1 that is observed upon pheromone stimulation *in vivo* [193]. During the mating response, a cell lacking Bni1 exhibits defects in the actin cytoskeleton, cell polarization, and cell fusion similar to those of a cell lacking Fus3; these phenotypes of a *fus3* cell can be substantially rescued by overexpression of Bni1 [193]. Bni1-promoted actin cables may be the avenue by which the Ste5 scaffold protein is delivered to the shmoo tip [194]. Thus, the formin Bni1 may be a primary target for the function of Fus3 in polarized growth and cell fusion, independent of the roles that Fus3 plays in imposing cell cycle arrest and inducing gene transcription.

4.3 Control of cell cycle progression

Pheromone stimulation leads to cell cycle arrest in the G1 phase in preparation for the formation of mating projections and eventual cell and nuclear fusion of the haploid partners. This cell cycle arrest is dependent upon a function of Far1 (Fig. 2) that is independent of its role in delivering the GEF (Cdc24) for GTP loading of Cdc42, which is, in turn, essential for both MAPK activation (via

the PAK, Ste20) [2] and cell polarization (via Bni1 and other effectors) [19]. Upon pheromone stimulation, Fus3 phosphorylates Far1 [195], which then is then able to associate with and inhibit the function of cyclin-CDK complexes (Cln1- and Cln2-bound Cdc28) [196, 197]. Whether this inhibitory effect is due, mechanistically, to direct inhibition of the catalytic activity of the Cdc28 CDK is controversial [198, 199]. Additionally, Fus3 and Kss1 can impose pheromone-induced cell cycle arrest in a Far1-independent manner, although the molecular basis for this effect seems to be indirect, namely via reducing expression of genes (*CLN1*, *CLN2* and *CLB5*) encoding cyclins necessary for the G1-S phase transition [47, 200].

Figure 2. Mechanisms of MAPK regulation of yeast cell cycle progression.



[Open in a new tab](#)

Fus3 (in response to pheromone) and Hog1 (in response to hyperosmotic stress) impose cell cycle arrest in the G1 phase via their direct phosphorylation of two different proteins

(Far1 and Sic1, respectively) that act as direct inhibitors of yeast CDK1 (Cdc28). Hog1 also imposes cell cycle arrest in the G2 phase via blocking the action of a protein kinase (Hsl1) necessary for initiating the ubiquitin- and proteasome-mediated destruction of a protein kinase, Swe1 (mammalian ortholog, Wee1), that is a specific antagonist of cyclin B (Clb)-bound CDK1. Slt2/Mpk1 (in response to cell wall stress) imposes G2 cell cycle arrest via inhibition (direct or indirect) of the phosphoprotein phosphatase, Mih1 (mammalian ortholog, Cdc25C), that is necessary to reverse the inhibitory tyrosine-specific phosphorylation installed on CDK1 by Swe1. See the text for further details.

Like pheromone stimulation, hyperosmotic stress also causes MAPK-mediated cell cycle arrest [201, 202]. Although this arrest is only transient, it seems important for osmoresistance. Unlike pheromone-imposed arrest, osmostress leads to cell cycle delays in both G1 and G2 [203, 204] (Fig. 2). Presumably stress responses are most efficiently and safely mounted when the cell genome is not in the vulnerable state of either replication or segregation (hence, either G1 or G2 arrest suffices). In contrast, mating specifically involves cells that must maintain their haploid genomic content, so only a G1 arrest is appropriate in this circumstance.

Timely passage from G1 to S requires the ubiquitin-dependent proteasome-mediated degradation of Sic1, a Cdk inhibitor (CKI) of S phase cyclin-Cdk complexes (Clb5- and Clb6-bound Cdc28) [205]. Upon hyperosmotic shock [201], Hog1 phosphorylates Sic1 at a position that reduces its ability to interact with a specificity subunit, the F-box protein, Cdc4 [206], of the ubiquitin ligase (E3), known as the Skp1-Cdc53/Cullin-F-box (SCF) complex, which mediates ubiquitinylation of Sic1 [207, 208]. Additionally, among the genes whose expression is reduced, rather than induced, when Hog1 translocates into the nucleus [92], are those encoding G1 cyclins (Cln1 and Cln2) [201]. This situation reduces the extent of Sic1 degradation because phosphorylation of Sic1 by G1 cyclin-bound Cdc28 is what marks it for recognition by Cdc4 [205]. Thus, Hog1 action stabilizes Sic1 through the combination of these two mechanisms, thereby stalling the G1-S transition.

The efficiency of passage from G2 to M is regulated, in part, by a morphogenesis checkpoint in which assembly of the septin collar at the bud neck leads to recruitment of an AMPK-related protein kinase, Hsl1. Hsl1 promotes entry into mitosis by recruiting and phosphorylating another protein, Hsl7, and together these factors act to stimulate degradation of Swe1, a protein kinase that phosphorylates and negatively regulates the M phase-specific B-type cyclin (Clb1 and Clb2)-

bound form of Cdc28 [209-211]. During osmostress, Hog1 reportedly phosphorylates Hsl1 at a site within its Hsl7-interacting domain, thereby preventing Hsl7 recruitment, thus stabilizing Swe1 and causing a delay in exiting G2 and entering M phase [202].

Perturbation of the actin cytoskeleton (for example, by exposure to the actin monomer-binding drug, latrunculin-B) activates Slt2/Mpk1 and causes an Slt2/Mpk1-dependent G2 arrest. Unlike Fus3- and Hog1-mediated regulation of the cell cycle via effects on CKIs (Far1 and Sic1, respectively), cell cycle arrest by Slt2/Mpk1 seems to occur via blocking the function of Mih1, the phosphatase that must act to reverse the inhibitory phosphorylation installed by Swe1 [212]. In the absence of Mih1 function, Swe1 action is sufficient to hold Clb-bound Cdc28 in check, preventing mitotic entry (Fig. 2). However, how Slt2/Mpk1 acts to prevent Mih1 function has not been determined at the molecular level.

In this section, we have enumerated mechanisms elucidated in yeast by which extracellular signal-activated or stress-induced MAPKs impose cell cycle arrest. Given the conservation of both MAPKs and cell cycle components across eukaryotes, some of these mechanisms may also be preserved in mammalian cells.

4.4 Transcription

Perhaps the most well-characterized function of MAPKs is their role in the regulation of gene expression at the transcriptional level (Fig. 3). Pheromone-regulated gene expression is dependent on the transcriptional transactivator, Ste12 [213-215]. Genes, such as *FUS1* [216] and *PRM1* [217], that are virtually not expressed in the absence of pheromone [215] contain multiple (≥ 3) tandem repeats of a *cis*-acting site (ATGAAACA), the pheromone response element (PRE), that is both necessary and sufficient to place a gene under the control of the pheromone response pathway [218-220]. Genes that are expressed at a significant basal level in the absence of pheromone, but further induced by mating pheromone (e.g. *STE2*, *MFA1* and *MFA2* in α cells, and *STE3* and *MFA1* in α cells) [218, 221-223], typically contain only 1–2 PREs, juxtaposed to the binding sites for other classes of DNA-binding transcription factors. Ste12 binds directly to the PRE in DNA via an N-terminal helix-turn-helix (HTH) motif related to, but divergent from, that found in classical homeodomain proteins [224]; Ste12 appears to be a member of the now-recognized winged-HTH family of transcriptional regulators [225]. The DNA-binding domain is situated within the first 164 residues of Ste12 (a 688-residue protein), followed by a homodimerization domain within the next ~50 residues, and then by a transcriptional activation segment (whose

minimum seems to be residues 301–335) [226].

Figure 3. Mechanisms of MAPK regulation of transcriptional initiation in yeast.



[Open in a new tab](#)

Inactive Kss1 resides mainly in the nucleus and acts as a transcriptional co-repressor by forming quaternary complexes with the heterodimeric transcription factor, Tec1-Ste12, and the repressors, Dig1 and Dig2. Ste7-dependent dual phosphorylation activates Kss1, permitting it to phosphorylate the Dig proteins and Ste12, thereby leading both to derepression and to activation of the transcription factor. Inactive Hog1 resides mainly in the cytosol and Pbs2-dependent dual phosphorylation activates Hog1 and promotes its translocation into the nucleus, where it stimulates transcription at some promoters, in part, by binding to and converting a transcriptional repressor, the Sko1-Cyc8/Ssn6-Tup1 complex, into a transcriptional activator, and, in part, by affecting the state of local

chromatin modification via recruitment of a specific histone deacetylase, the Sin3-Rpd3 complex. Active Hog1 can also stimulate transcription at other promoters by phosphorylating and binding to a transcriptional activator, such as Hot1, and thereby serving as an adaptor or mediator that also binds to and recruits RNA polymerase II holoenzyme. See the text for additional details.

At the promoters of genes, like *FUS1*, in which the tandem PREs have the appropriate spacing, Ste12 binds as a homodimer [224]. However, at other promoters, Ste12 binds to DNA as a hetero-oligomeric complex with other transcription factors, such as an Ste12-(Mcm1)₂ complex at the promoters of certain a cell-specific genes [218, 223], an Ste12-Mata1-(Mcm1)₂ ternary complex at the promoters of certain a cell-specific genes [221, 222], and an Ste12-Kar4 complex at the promoters of genes expressed late in the mating process, like *KAR3* (which encodes a kinesin involved in karyogamy) [227]. At the *PRY3* promoter, Ste12 binding is even able to shift which TATA sequence is the preferred site for transcription initiation, leading to the generation of a transcript that is 452 shorter than the mRNA made in the absence of pheromone stimulation [228].

Mcm1 is the yeast ortholog of a ubiquitous mammalian transcriptional activator, serum response factor (SRF) [229], Mata1 is a homeobox-containing transcription factor [230, 231], and Kar4 is a transcription factor whose expression requires Ste12 and is pheromone-inducible [232]. Exposure to pheromone elevates the expression of a large number of genes significantly, including those encoding proteins required for cell-cell recognition and cell-cell fusion, components of the pathway itself (positive feedback), and factors that down-regulate the pathway (negative feedback) [215, 233, 234]. Fus3 action stimulates Ste12-dependent gene expression by phosphorylating Ste12 itself [195, 235, 236], but mainly by phosphorylating and relieving repression by two Ste12-binding repressors, Dig1 and Dig2 [237, 238].

Expression of genes required for invasive growth in haploids also requires Ste12, but in this case, Ste12 does not usually bind directly to DNA, but does so primarily via protein-protein association with yet another, dimeric, DNA-binding transcription factor, Tec1 [239-241]. Like Ste12, Tec1 can also function as a transcription factor on its own [242]; however, the site where a composite Ste12-(Tec1)₂ complex binds is referred to as a filamentation response element (FRE) [234, 241] (regulation of Ste12 specificity during different signal responses is discussed further in Section 5.4.2). Two promoters that serve as models for nutrient-responsive gene regulation are that of

MUC1/FLO11 [243], encoding a filamentous growth-specific mucin-like flocculin/adhesin, and *STA1* [244], which encodes a secreted glucoamylase. These promoters integrate signals from different pathways through consensus binding sites for multiple transcription factors, including Ste12-(Tec1)₂ (downstream of the MAPK Kss1), Flo8 (downstream of the PKA Tpk2), Msn1 (downstream of the AMPK Snf1), and a Flo8-like transcription factor, Mss11 [245, 246]. At the *STA1* promoter, a sequential integration model for activation has been proposed in which Ste12-(Tec1)₂ binding to a FRE recruits the Swi/Snf remodeling complex, which facilitates the cooperative interaction of Flo8-Mss11 with the promoter, leading to RNA polymerase II recruitment and transcriptional initiation [244].

These transcriptional activators are all regulated by competition with transcriptional repressors. At the *MUC1/FLO11* promoter, the repressors Nrg1 and Nrg2 inhibit the binding of Msn1 until Snf1-mediated phosphorylation lifts Nrg1- and Nrg2-imposed repression [80]; likewise, Tpk2-mediated phosphorylation displaces a repressor, Sfl1, thereby permitting binding of Flo8 (and its co-regulator, Mss11) [247]; similarly, Kss1-catalyzed phosphorylation and displacement of Dig1 and Dig2 permits stimulation of transcription by Tec1-tethered Ste12 [237-239]. Collectively, rather similar events also occur at the *STA1* promoter [248].

At pheromone-responsive promoters, Ste12 homodimers associate with both Dig1 and Dig2, which are only weakly related (27% identity, 35% similarity). In one study, which utilized biochemical pull-down assays, the larger Dig1 (452 residues) bound to the C-terminal region of Ste12 (residues 262-to-594 of Ste12 were sufficient for this interaction), whereas the smaller Dig2 (323 residues) bound to the N-terminal DNA-binding domain of Ste12 [249]. However, in another study, the minimal transcriptional transactivation segment of Ste12 (residues 301–335) was sufficient for interaction with either Dig1 or Dig2, at least as judged by the two-hybrid method [226]. At the promoters for genes involved in filamentous growth, only Dig1 purportedly associates with the Ste12 in Tec1-containing complexes [239]. This situation arises, allegedly, because Tec1 associates with the DNA-binding domain of Ste12, which appeared, in the first study cited above [249], to also be the site where Dig2 binds; and, hence, the two proteins would be expected to compete with each other [239]. However, this claim is not supported by the fact that deletion of both *DIG1* and *DIG2* is required to derepress filamentation genes [237, 238] and by the other study cited above [226], which reported that both Dig1 and Dig2 bind to a similar region of Ste12.

In any event, once phosphorylated, Ste12 binds Dig1 and Dig2 more weakly [237, 238]. Moreover, at least at FREs, Kss1 is also present in the Dig1- and Dig2-containing repressed complexes [250]. Inactive Kss1 binds directly to and contributes to repression of Ste12 [251]. Phosphorylation of

Kss1 by its upstream MAPKK (Ste7) simultaneously weakens Kss1-Ste12 interaction and activates the catalytic activity of Kss1 [251]. Activated Kss1, in turn, phosphorylates Dig1, Dig2, and Ste12 [237, 238, 251], thereby leading to full derepression of Ste12-dependent expression. Kss1 also binds directly and tightly to both Dig1 and Dig2 [237, 238, 250, 251]. Collectively, these data suggest that the inactive state of Kss1 serves as a transcriptional co-repressor, that activation of Kss1 weakens repression, and that, once activated, Kss1-mediated phosphorylation of Dig-Ste12-(Tec1)₂ complexes remodels them appropriately to promote transcriptional activation by Tec1-tethered Ste12 (Fig. 3). Consistent with this view, like Dig1 and Dig2, the bulk of Kss1 is always found in the nucleus before or after pathway stimulation (L. Shiow, J.X. Zhu-Shimoni, R.E. Chen and J. Thorner, unpublished results), as noted before for overexpressed Kss1 [37].

As already mentioned briefly earlier, activated Slt2/Mpk1, the MAPK of the CWI pathway, is also a regulator of gene expression [107, 139, 252], primarily via the direct phosphorylation and activation of the transcription factor, Rlm1 [139-141, 253]. Genes with Rlm1-binding sites in their promoters are enriched for those encoding proteins and enzymes involved in cell wall structure or biogenesis [139, 254, 255]. It has also been reported that Slt2/Mpk1 interacts physically with Swi4 (a subunit of the heterodimeric Swi4-Swi6 transcription factor, termed SBF), that recruitment of Swi4 to promoters is reduced in strains lacking Slt2/Mpk1, and that Slt2/Mpk1 and Swi4 share a set of target genes that are independent of Swi6, including the Pho85/CDK5-specific cyclin, Pcl1, and the 1,3- β -glucan synthase, Gsc2 [106, 256]. These findings have led to the proposition that Slt2/Mpk1 may be involved in a novel Swi4-mediated (but Swi6-independent) mode of gene regulation, but the precise mechanism of this gene control is unclear [256]. Finally, another interesting direct substrate of Slt2/Mpk1 is Sir3, a protein required for the maintenance and spreading of heterochromatin [257]. Mutation of the Slt2/Mpk1 phosphorylation site on Sir3 increases yeast lifespan an average of 38% [258]. Sir3 is also reportedly a substrate for Fus3 [259]. These observations suggest molecular connections between the sensing of extracellular conditions, gene silencing, and cellular senescence.

In response to hyperosmotic stress, Hog1 translocates from the cytosol into the nucleus where it affects the expression of a large number of genes [92, 260, 261], as mentioned earlier. Unlike action of Fus3 and Kss1 through Ste12, and Slt2/Mpk1 largely through Rlm1, Hog1 influences the expression of genes driven by a wide variety of transcription factors (those grouped together are sequence-related), including: Hot1 and Msn1 (activators); Msn2 and Msn4 (activators); Sko1 (repressor); and Smp1 (activator; very related to Rlm1, but similarity mainly confined to their N-terminal MADS box-type DNA-binding domains)[104, 260, 262-266]. All of these factors interact with Hog1 at the promoters of the respective target genes, and the role of Hog1 at some of them

has been reasonably well characterized ([Fig. 3](#)). Phosphorylation of Smp1 by Hog1 is required for its activator function [[104](#)]. In the case of Sko1, this DNA-binding protein acts as a repressor by binding the Tup1-Ssn6/Cyc8 complex, which, in turn, recruits the ISW2 chromatin-remodeling and nucleosome-positioning complex and several histone deacetylases, including Hda1 [[267-269](#)], and thereby prevents RNA polymerase II access to chromatin. Hog1-mediated phosphorylation of Sko1 somehow converts it into an activator, perhaps by causing it to jettison some or all of its repressive co-factors [[270](#), [271](#)]. How Hog1 action stimulates Hot1 function is different still. Although Hog1 phosphorylates Hot1, this modification is not necessary for the activator function of Hot1; rather, active Hog1 is required for Hot1 binding at some promoters [[272](#)], and it is active Hog1 that serves as an adaptor to recruit RNA polymerase II to the promoter-bound Hog1-Hot1 complex [[273](#)]. Curiously, and contrary to the dogma that histone deacetylation is correlated with repression of gene expression, at many osmo-responsive promoters, it has been reported that Hog1 interacts with and thereby recruits another class of histone deacetylase, the Sin3-Rpd3 complex; but, at these genes, Rpd3 action somehow enhances, rather than prevents, recruitment of RNA polymerase II, thus promoting osmostress-induced gene expression by yet another mechanism [[274](#)].

Two different groups [[275](#), [276](#)] have recently reported that, beyond interacting with regulators of transcription initiation, Hog1 also interacts with transcription elongation factors. Furthermore, these studies also claim that Hog1 can be found bound all along the coding regions of osmo-responsive genes and that Hog1 can enhance mRNA synthesis of normally osmo-responsive genes, even when the promoter has been rendered Hog1-insensitive (by replacement of the native control elements with a bacterial LexA operator driven by the artificial constitutive activator, LexA-VP16) [[275](#), [276](#)]. Thus, the Hog1 MAPK may function not only as an integral component of several different types of transcriptional initiation complexes, but may also serve as a co-factor for transcriptional elongation.

In summary, MAPKs regulate transcription not only by directly phosphorylating transcription factors, but also by participating structurally as stable components of complexes that can serve as either transcriptional repressors or activators. This scenario may be true not only for MAPKs, but also for other families of kinases [[275](#), [277](#)]. The function of a MAPK-containing complex as a repressor or activator, or its structural integrity, can be altered upon activation of the MAPK or by the phosphorylation of components by the active MAPK. Additionally, it seems that MAPKs can interact physically with transcriptional regulators other than promoter-binding transcription factors, such as histone deacetylases, transcriptional elongation factors, and even RNA polymerase II itself. The mechanisms by which MAPKs regulate transcription constitute an unexpectedly

diverse repertoire for directly connecting the outputs of signal transduction pathways to the primary gene regulation machinery of the cell.

4.5 Translation

In addition to the many characterized instances in which MAPKs regulate mRNA synthesis, it has recently become apparent that MAPKs also perform important functions in the regulation of gene expression at the level of protein synthesis. Hyperosmotic shock induces a transient decrease in the rate of translation [278], which recovers as the cells adapt to the stress [279]. This effect is thought to be mediated by Hog1, which phosphorylates Rck2 [103], a MAPKAP kinase [280] that is distantly related to the type II Ca^{2+} /calmodulin-dependent protein kinase (CaMKII) subfamily (although Rck2 does not bind Cmd1 (yeast calmodulin) nor is its activity affected by Cmd1 [281]). With regard to the effects observed on translation upon activation of Hog1, it is interesting that Rck2 is able to phosphorylate *in vitro* yeast translation elongation factor 2 (EF-2; encoded by the *EFT1* and *EFT2* genes) [103, 281]. Phosphorylation of the mammalian homolog causes its dissociation from ribosomes and stimulates protein synthesis [282]. This effect of phosphorylation is more consistent with a role for Rck2 in promoting recovery from the initial translational repression observed in response to hyperosmotic stress. In this regard, one study does report that Hog1 is required not for the initial decrease in translation, but rather for the subsequent recovery [279].

In a similar manner, it has been demonstrated that treatment of cells with mating pheromone dramatically affects what transcripts are found in association with polysomes and does not correlate well with the relative abundance of those transcripts globally [283-285]. It seems likely, therefore, that MAPKs differentially influence which transcripts are selected for translation and the efficiency with which they are translated. This discrimination could be achieved, for example, by MAPK-mediated inhibition of bulk translation in a manner that affects most, but not all, transcripts. Such an effect has been observed upon the stress of glucose withdrawal [286]. Alternatively, MAPKs could facilitate the selective protection of specific transcripts from some constitutive mechanism, like mRNA turnover. There is at least claim of a MAPK-dependent translational regulation that occurs at the level of a target transcript, as opposed to the level of a general translation factor. It has been reported that a very unstable mRNA, *MFA2* (one of the genes that encodes α -factor pheromone) [287], displays a marked increase in its translation when cells are shifted from glucose (a fermentable carbon source) to glycerol (a non-fermentable carbon source) and that this response requires the MAPK, Hog1 [288]. The *MFA2* mRNA possesses AU-

rich elements (AREs) in its 3'-untranslated region (3'-UTR) just upstream of its poly(A) tail, and during growth in glucose, the 3'-UTR is occluded by spreading of the poly(A)-binding protein, Pab1 [288]. During growth in glycerol, full-length Pab1 is replaced by a truncated form of Pab1, and *MFA2* translation becomes derepressed [288]. Likewise, Pab1 binding and *MFA2* repression do not occur in cells lacking Pub1, an RNA-binding protein that recognizes the AREs and helps recruit Pab1. However, the mechanisms underlying these carbon source-elicited Hog1-dependent effects are obscure, to say the least. First, these same workers have reported that loss of Pub1 generally destabilizes the transcripts that bind Pub1 [289], contrary to what they purportedly observed with *MFA2* mRNA. Second, binding of Pab1 to the poly(A) tail generally promotes cap-dependent translation of mRNAs, through the mutually reinforcing associations of the eIF4G scaffold protein with both the 7-methyl G cap-binding protein (eIF4E) and Pab1 [290], again contrary to what was reportedly observed for the *MFA2* transcript.

Another recent study suggests that translational regulation may be involved in setting the level of Ste12 made during the filamentous growth response [291]. Under nutrient limiting conditions, the amount of Ste12 increases, while abundance of *STE12* mRNA does not. This apparent discrepancy is explained, at least in part, by the fact that *STE12* mRNA becomes enriched in the polyribosome fraction under these conditions. Deletion of either Caf20, which is a yeast eIF4E-binding protein (4E-BP), or Dhh1, a DEAD-box RNA helicase found in cytoplasmic mRNA-processing centers called P-bodies [292], prevents the increase in Ste12 protein without affecting mRNA levels. Consistent with this, cells lacking Caf20 or Dhh1 exhibit observable defects in filamentous growth [291]. Again, these reported effects are somewhat counter-intuitive. Loss of 4E-BP should stimulate cap- and eIF4E-dependent translation and loss of Dhh1 should spare an mRNA from delivery to the P-body and thus increase the amount of the transcript available for translation. Nonetheless, at least one other group has also reported that loss of 4E-BPs impedes filamentous growth in yeast [293], suggesting that without 4E-BPs to restrict the function of eIF4E and presumably cap-dependent translation, the mRNAs needed for this developmental response cannot be selectively translated. This notion, in turn, suggests that perhaps the mRNAs for such factors are translated preferentially via internal ribosome entry sequences (IRES's). Indeed, this possibility is consistent with the fact that transcripts for many genes involved in filamentous growth (and other stress responses) have unusually long 5'-UTR's and do seem to contain functional IRES elements (W. Gilbert and J. Doudna, personal communication). Nonetheless, it is not known whether any of these phenomena are dependent on the action of any MAPK; however, at least in the filamentous growth pathway, a primary candidate might be Kss1, which is already known to regulate Ste12 function by several *post*-translational mechanisms (see Section 4.4), although Fus3 and other kinases also modulate the level and thus the function of Ste12 *post*-

translationally [294, 295].

5. Regulation of MAPK signal transduction

5.1 MAPK scaffolds.

It is noteworthy that three of the five MAPK pathways discussed in this article (Fig. 1) depend on proteins that serve an adaptor, anchoring, or scaffold function to ensure that critical components of the MAPK cascade can be readily engaged by the correct upstream initiating signal and are held in close proximity to each other at the plasma membrane. Presumably this latter kind of enforced intimacy is necessary to maximize signal propagation while minimizing the risk of inadvertently activating inappropriate and/or unproductive responses. In the pheromone response pathway, Ste5 binds the G $\beta\gamma$ (Ste4-Ste18) released from pheromone receptors, as well as the MAPKKK Ste11, the MAPKK Ste7, and the MAPK Fus3 of the pathway [10-12]. In the hyperosmotic stress response, Pbs2 serves a similar scaffold function (even though it is itself the dedicated MAPKK of the HOG pathway) because it binds the integral membrane protein that is a component of one of the primary osmosensors (Sho1), its immediate upstream activators (the three MAPKKKs, Ste11, Ssk2 and Ssk22), and its target MAPK (Hog1) [9, 93, 95]. In the CWI pathway, the Spa2 scaffold protein, which interacts with Bni1, Bud6 and other plasma membrane-localized actin-associated proteins, as well as with other components of the polarisome required for polarized growth, also binds two MAPKKs, Mkk1 and Mkk2, as well as their target, the Slt2/Mpk1 MAPK [296, 297]. The need to make all of these contacts perhaps explains the very large size (1466 residues) of the Spa2 polypeptide. The lack of sequence relatedness among these scaffolds suggests that other, as yet unidentified scaffolds, will be difficult to predict by any kind of simple computational analysis.

Ste5 was the first MAPK scaffold protein identified in any organism and is arguably the best characterized [298, 299]. Recent findings demonstrate that Ste5 serves as much more than a passive platform on which to simply moor the constituent kinases of the pheromone response pathway. In vegetatively growing cells, Ste5 undergoes rapid nucleocytoplasmic shuttling, with a predominantly nuclear distribution in naïve cells. Upon pheromone stimulation, a significant proportion of the population of Ste5 molecules localizes to the tip of the mating projection [300-302]. Ste5 binds to G $\beta\gamma$ through its RING-H2 domain [13, 28, 29], but also associates with acidic phosphoinositides at the plasma membrane through an N-terminal basic amphipathic helix (PM motif) [32] and with PtdIns4,5P₂ specifically via an internal PH domain [33]. Each of these three interactions individually, or even any two of them, is insufficient to stably tether Ste5 to the

plasma membrane; however, all three acting in conjunction are sufficient to do so [33]. Furthermore, studies suggest that G β γ -binding may induce a conformational change in Ste5 that perhaps optimizes the orientation of the bound kinases [303] or exposes the PM and PH domains to permit efficient membrane binding [33], or both. However, the converse may be true, namely that association with G β γ is passive whereas the ensuing association with membranes is what causes conformational change that is important for Ste5 function, because it has been observed that artificial tethering of Ste5 to intracellular membranes amplifies signal transduction downstream of constitutively-active (Ste20-independent) alleles of the Ste11 MAPKKK [304]. Other studies suggest that proper pheromone-stimulated localization of Ste5 from the nucleus to the cell cortex depends on its movement along actin cables assembled in a Bni-dependent manner [194], as mentioned earlier, and on direct interactions between Ste5 and Cdc24 (Cdc42 GEF) [305]. Although the former is supported by independent findings from another group [193], that latter proposal is most likely incorrect since an allele of Cdc24 (*cdc24-4*) that cannot interact with Ste5 does not exhibit any detectable mating defect [306].

Ste5 is found in what appear to be large complexes [10]. Moreover, efficient interallelic complementation is observed between *ste5* alleles that are individually non-functional [307, 308]. These observations suggest that Ste5 monomers can self-associate to form an oligomer and that Ste5 can function to activate the pheromone response pathway only when it is in the oligomeric state. Indeed, fusion of a heterologous dimerization domain (glutathione S-transferase) rescues the mating defect of non-oligomerizing *ste5* alleles [28] and promotes nuclear export and membrane recruitment of Ste5 [309]. Oligomerization of Ste5 seems to require its RING-H2 domain [28, 308, 309]. Additionally, the N- and C-terminal portions of Ste5 can interact *in vitro* [303]. Together, these and other data suggest, first, that only properly oligomerized and membrane-localized Ste5 can present its bound cargo of MAPKKK Ste11 to its membrane-tethered upstream activator, the PAK Ste20, in the appropriate manner and, second, that Ste5 oligomerization also serves to arrange the interrelationships between its other bound kinases so as to optimize signal propagation between them. Remarkably, it is still not known whether the active Ste5 oligomers are dimers or higher order structures, whether oligomerization occurs in a parallel or antiparallel orientation, or whether the kinases of the MAPK cascade phosphorylate each other *in cis* on a single scaffold molecule or *in trans* across the monomers within an oligomer.

Recently, it has been observed that, *in vitro*, Ste5 binds Fus3 directly (but not Kss1) [310], a finding consistent with the fact that Fus3 is readily detectable in immunoprecipitates of Ste5 isolated from cell extracts, but Kss1 is not [303]. Second, it has been observed that a peptide fragment of Ste5 corresponding to the site where Fus3 docks acts as an apparent allosteric

activator *in vitro* by stimulating Fus3 auto-phosphorylation on the Tyr of the -TEY- motif in its activation loop [310]. This form is more active than unphosphorylated Fus3, but significantly less active than dually-phosphorylated Fus3. The physiological significance of this finding is unclear for several reasons. *In vivo*, it has been shown previously that both Tyr and Thr phosphorylation depend on Ste7 [311], and there is no evidence to show that this allosteric autoactivation of Fus3 can occur in the context of full-length Ste5 or when Ste7 is present, or be preserved in the face of competing cellular phosphatases. Nonetheless, the observation is intriguing because it provides yet another indication that Ste5 has a role in actively establishing the conformational state and perhaps activation state of at least one of its bound kinases.

It has been suggested [302] that another role for Ste5 is to permit signal amplification in this MAPK pathway. This idea is based on analysis of the dynamics of green fluorescent protein (GFP)-tagged versions of several of the pathway components determined by fluorescence recovery after photobleaching (FRAP). Compared to Ste5, whose localization at the plasma membrane seems quite stable by this measure, Fus3 displays high rates of association and dissociation. Theoretically, this situation would allow each scaffold molecule to participate in many rounds of activation and thus produce multiple molecules of activated MAPK, which are then free to translocate into the nucleus or diffuse to other effector sites without being complexed with Ste5. It has also been observed that upon pheromone stimulation, Ste5 is phosphorylated by the mating pathway kinases [312], and such modifications may assist in dissociating bound components, like Fus3. If so, these phosphorylations would also need to be very dynamic to support the amplification scenario just described. Also, the cellular and cortical abundance of Ste5 increases in a MAPK-dependent manner [313], indicating some sort of positive feedback loop between the scaffold and the MAPKs.

5.2 Subcellular localization

Spatial heterogeneity and compartmentalization within cells permits pathways and molecular components to be regulated on the basis of their subcellular localization and molecular accessibility. The localized recruitment in space and time of many factors (for example, the PAK, Ste20, or the scaffolds, Ste5, Pbs2 and Spa2) is an integral part of the operation of the MAPK pathways in which they are found (see Sections 2.2, 3, 5.1 and 5.4.1). For additional viewpoints on the physiological consequences of localization of pathway components, see [87, 302, 314]. Here we focus primarily on regulation of the subcellular distribution of the MAPKs themselves.

In mammalian cells, the MAPK Erk2 is localized in the cytosol when inactive and enters the nucleus after activation [315, 316]. The upstream MAPKKs, Mek1 and Mek2, have a high-affinity docking site for Erk2, but also contain a potent nuclear export signal (NES), thus ensuring that prior to activation, the complex is in the cytoplasm. Upon activation, dually-phosphorylated Erk2 dissociates from Mek1 and Mek2 and enters the nucleus by a combination of (a) passive diffusion as a monomer, (b) Ran and importin β -dependent active transport as a dimer, and (c) Ran/importin β -independent facilitated diffusion via direct interactions with the nuclear pore complex [317-321]. In certain cell types, active Erk2 is sequestered in the cytoplasm by the Golgi-localized transmembrane protein, Sef, or the small death effector domain (DED)-containing protein Pea-15 [322, 323], thereby channeling the signal exclusively to cytoplasmic effectors. Mek1 and Mek2, which undergoes nucleocytoplasmic shuttling, can retrieve Erk2 from the nucleus once it has undergone inactivation/dephosphorylation.

In yeast, the MAPK Hog1 displays a localization pattern similar to that of Erk2, namely cytoplasmic when inactive, and then nuclear after activation, but transiently. Nuclear translocation of Hog1 upon activation is very rapid (within 5 min) and depends its dual phosphorylation [324]. Moreover, nuclear import requires that Hog1 itself be an active kinase [325] and is mediated by the karyopherin (importin β family member), Nmd5 [100]. Contrary to a previous claim [324], subsequent export of Hog1 does not require its kinase activity [325], but it does require Xpo1/Crm1, another importin β family member [100]. However, specific NLS or NES signals have not yet been identified in Hog1. Hog1 localization is also influenced by relatively high-affinity interaction partners in each compartment. Of particular note, in the nucleus, Hog1 interacts with many transcriptional regulators (see Section 4.4) and a protein-tyrosine phosphatase, Ptp2, which is responsible for Hog1 deactivation in that compartment. In the cytosol, Hog1 associates with its MAPKK, Pbs2, and with another protein-tyrosine phosphatase, Ptp3, which tethers it in the cytoplasm [326].

Unlike Erk2 and Hog1, the MAPKs Fus3 and Kss1 are predominantly nuclear (although also observable in the cytoplasm) both when active and inactive [37, 314]. Although the rate of Fus3 nuclear entry, as assayed by FRAP, is not significantly altered by pheromone treatment [302], a small increase in the nuclear/cytoplasmic Fus3 ratio has been reported [327]. Additionally, a population of highly mobile Fus3 accumulates at the tips of cellular mating projections formed upon pheromone stimulation [302]. Artificial tethering of the scaffold Ste5, but not a mutant version lacking the Fus3-binding region, all around the plasma membrane was sufficient to recruit Fus3, as well as unphosphorylatable and catalytically-inactive alleles, to the plasma membrane, suggesting that localization of Fus3 to the site of polarized growth during pheromone response

may be Ste5-mediated [302]. Consistent with this view, cortical recruitment of Fus3 is not observed in cells lacking Ste5 [302]; however, the lack of formation of mating projections in such cells (due to the absence of downstream signaling) means that one cannot exclude the formal possibility that recruitment of Fus3 to the shmoo tip normally requires the events and processes required for polarized morphogenesis itself or is mediated by other factors recruited into the mating projection other than the Ste5 scaffold itself. In this regard, it is noteworthy that pheromone-induced recruitment of both Fus3 and Ste5 to the cell cortex is absent in cells lacking the formin Bni1 [194]; but again, the complication in interpreting this result unequivocally from a mechanistic standpoint arises from the fact that Bni1-deficient cells also do not form mating projections. Altogether, therefore, the requirement of pathway activity for mating projection formation has so far precluded a definitive determination of whether recruitment of Fus3 to the mating projection tip is mainly due to its interactions with upstream activators or to provide an opportunity to interact with downstream targets/effectors.

A claim that a significant amount of Fus3 is associated with Gpa1 (Gα) [328] is unlikely to be of biological relevance. In fact, even when Gpa1 or Msg5 (a dual-specificity, i.e. Ser/Thr- and Tyr-specific, protein phosphatase that can deactivate Fus3) is over-expressed, there is only a very slight decrease in the nuclear/cytoplasmic ratio of Fus3 after pheromone stimulation. Furthermore, although these subtle effects on the nucleo-cytoplasmic distribution of Fus3 were reportedly dependent on the karyopherin, Kap104 [327], deletion of the *KAP104* gene, which is not an essential gene, had no effect on the Fus3 distribution in otherwise wild-type cells [327].

Thus, although it seems that Fus3 and Kss1 constitutively and rapidly shuttle between the nucleus and cytoplasm independently of their activation state or catalytic activity [302], it is still unclear what the major determinants are for establishing the observed steady-state distribution wherein the bulk of these MAPKs are nuclear. As for Hog1, no classical NLS or NES sequences have been identified in Fus3 or Kss1. It may be the case that regulation of the subcellular distributions of Fus3 and Kss1, is carried out in a manner more similar to that of Erk2 than of Hog1, occurring not at the level of energy-dependent translocation, but primarily at the level of tethering and release from partners that are themselves independently compartmentalized. Unlike Erk2, however, the dominant “anchors” for Fus3 and Kss1 in unstimulated cells must be nuclear, not cytoplasmic. Perhaps the most obvious candidates might be transcriptional regulators, as Fus3 and Kss1 have been shown to interact physically with the transcription factors Dig1, Dig2, and Ste12 [237, 251], and to be present in complexes bound to the DNA of several genes [275]. However, at least Fus3-GFP is still predominantly nuclear even in *dig1Δ dig2Δ ste12Δ* cells [302], and we have observed the same for Kss1-GFP (R.E. Chen and J. Thorner, unpublished results).

5.3 Temporal characteristics and pathway inactivation

The responses of cells to external signals and acute stresses, while necessary for viability or developmental transitions in the short term, often involve behaviors that can impair growth or viability in the long term. Accordingly, MAPKs, particularly those that either arrest or substantially divert resources away from cellular proliferation, generally are kept at low activation levels in the absence of their appropriate signals and are rapidly inactivated following the course of a relatively brief response. For example, Fus3 and Hog1, both of which induce cell cycle arrest, are both inactivated within an hour or less after their activation, permitting cells to recover and resume vegetative growth (as either haploids or diploids, in the case of the pheromone response, depending on the success of mating). Tellingly, artificially prolonging the activation of either MAPK causes inviability. Inactivation of MAPK pathways occurs through both constitutive and induced (negative feedback) mechanisms. In addition to their gross roles in preventing inviability, however, it is also becoming clear that mechanisms for inactivating MAPK pathways are also well-suited, and utilized, for more finely tuned regulation that affects the nature of the quantitative profile of signal transduction activity spatially and temporally. Modulation of the speed, magnitude, and duration of pathway activation can play a role in determining the qualitative nature of the output elicited [[43](#), [329-331](#)].

Because MAPKs are only fully active when phosphorylated on both the Thr and Tyr in their activation loop, one effective mechanism of MAPK inactivation is dephosphorylation by protein phosphatases of the serine/threonine, tyrosine, or dual-specificity classes. For other recent reviews on the roles of phosphatases in MAPK signaling, see [[332](#), [333](#)]. Fus3 has been shown to be dephosphorylated by the dual-specificity phosphatase Msg5 and the protein-tyrosine phosphatases Ptp2 and Ptp3; Hog1 by Ptp2, Ptp3 and the serine-threonine phosphatases Ptc1, Ptc2 and Ptc3; and Slt2/Mpk1 by Msg5, Ptp2, Ptp3 and the dual-specificity phosphatase Sdp1 [[138](#), [333-344](#)]. Although some of the same phosphatases target different MAPKs, substrate preferences vary, dictated by differential binding affinity and/or subcellular localization. Another mechanism operates in the case of a PP2C, Ptc1, that acts on Hog1. Ptc1 binds to an adaptor protein, Nbp2, which contains an internal SH3 domain that preferentially associates with a PxxP motif in the N-terminal regulatory domain of Pbs2, at a site distinct from the PxxP motif by which the SH3 domain at the C-terminus of Sho1 binds to Pbs2. Therefore, presumably, Pbs2 can interact simultaneously with both positive and negative regulators of the HOG MAPK pathway [[345](#)]. Since Pbs2 acts as a scaffold to bind Hog1, the association of Nbp2 with both Ptc1 and Pbs2 delivers the phosphatase to its MAPK target, Hog1. The specific phosphatase(s) that act to down-regulate Kss1 have not yet been identified.

As assayed by the level of phosphorylation in a given MAPK in cells lacking particular phosphatases, different phosphatases contribute to different extents and with different efficiencies with respect to their effects on the unstimulated, stimulated, and adapted states of their target MAPKs [333]. The temporal relationships observed are due, in part, to additional regulatory interactions between the MAPK pathway and the phosphatases. For example, *Msg5* has a significant role in down-regulating the activity of *Fus3* to promote recovery from pheromone stimulation. For *Slt2/Mpk1*, however, *Msg5* seems more important for maintaining the low basal phosphorylation of *Slt2/Mpk1* and is not much involved in down-regulating *Slt2/Mpk1* after its stress-induced activation. Consistent with these roles, it is not surprising that the *MSG5* gene is expressed at a significant level under all conditions and is transcriptionally induced upon pheromone stimulation [334] but not upon heat shock or oxidative stress [343]. Furthermore, it has been observed that, during cell wall stress, the affinity of *Msg5* for *Slt2/Mpk1* decreases [344], and *Slt2/Mpk1* phosphorylates *Msg5* (although the role of this modification in the observed reduction of affinity was not explored) [344]. Similarly, expression of *PTP2* and *PTP3* is induced in a *Hog1*-dependent manner, as expected if these phosphatases are part of the negative feedback loop that down-regulates activated *Hog1* [336].

Besides dephosphorylation of MAPKs, MAPK pathway activity can also be attenuated in a timed manner by the ubiquitin-mediated degradation of pathway components. The unphosphorylated form of *Ssk1*, which promotes HOG pathway activation via the *Sln1* branch, is degraded more rapidly than phosphorylated *Ssk1*, promoting recovery from osmostress [346]. In the pheromone response pathway, degradation of the MAPKK, *Ste7*, which is dependent on its phosphorylation by its activating MAPKKK, *Ste11*, is thought to contribute to recovery from pheromone-induced cell cycle arrest [347, 348]. Similarly, prolonged exposure to pheromone eventually leads to the *Fus3*-dependent degradation of the downstream transcription factor, *Ste12* [294]. The pheromone response pathway *Gα* (*Gpa1*) seems to undergo ubiquitin-mediated degradation [349, 350], although whether or not this is regulated by pathway activity is unclear. In a variant mechanism for potential degradation-controlled pathway regulation, *Ste11* seems to undergo rapid cycles of synthesis and destruction, although its total levels remain constant during pheromone stimulation [351], so the biological significance of this observed flux is not immediately obvious. On the other hand, in contrast to cases wherein pathway activity-induced degradation acts as a timer that limits the duration of activation, one could view this rapid turnover and replenishment as a means to ensure swift pathway shutoff upon cessation of upstream input, since the newly made *Ste11* will not become activated (phosphorylated) when there is no active *Ste20* to do so. However, this mode of *Ste11* regulation does not seem to occur during osmostress [351]. Hence, *Ste11* tethered to *Ste5* is somehow more susceptible to this stimulus-induced degradation than *Ste11* tethered to

Sho1, Pbs2, and Ste50.

Down-regulation of G protein-initiated signaling upstream of many of the MAPK pathways occurs via action of the cognate GAPs. In the case of heterotrimeric G proteins, the protein responsible for stimulating GTP hydrolysis bound to a G α subunit is a member of the family of Regulators of G protein Signaling (RGS) proteins. The first such protein recognized in any organism and the primary negative regulator of the pheromone response pathway is the prototype RGS protein, Sst2 [352, 353]. Occupancy of a pheromone receptor by its ligand causes a conformational change that allows the receptor to act as a GEF on Gpa1; simultaneously, this conformational change also exposes the C-terminal tail of the receptor, permitting the binding of Sst2 to the cytoplasmic tail of the receptor via two DEP domains situated in the N-terminal half of Sst2 [354]. Once tethered at the membrane, the C-terminal RGS domain of Sst2 is now in close proximity to its substrate, GTP-bound Gpa1, and can act on it efficiently. This situation ensures that pheromone response in the cell will only occur if the amount of pheromone to which it is exposed is high enough to activate enough receptors to generate more Gpa1-GTP than the Sst2 RGS protein can handle. Also, as expected for a classical negative feedback loop, the *SST2* gene is expressed at a significant basal level, but is also induced in response to pheromone, and the amount of Sst2 protein increases 10–20 fold [352]. It has been variously reported that the DEP domain-containing region of Sst2 associates with, on the one hand, a member of the Pumilio family (Puf) repeat class of mRNA-binding proteins, Mpt5/Puf, and that this interaction somehow contributes to recovery from pheromone response in an RGS-independent manner [355, 356] and, on the other, with several proteins (e.g. Vps36, Pep12/Vps6 and Tlg2) involved, directly or indirectly, with trafficking to the endosome and/or vacuole [357]. Since Sst2 associates tightly with pheromone receptors [354] and the receptors undergo robust constitutive and ligand-induced endocytosis in a ubiquitin-dependent manner [358], the latter interaction with proteins involved in endosomal trafficking may be physiologically significant.

Other instances of negative regulation in MAPK pathways have been identified, but are less well characterized. Phosphorylation by Fus3 of its scaffold protein, Ste5, down-modulates pheromone pathway output by an unknown mechanism [310]. Quite recently, it has been shown that the G1 cyclin-bound form of Cdc28 multiply phosphorylates Ste5 and that these modifications prevent efficient association of Ste5 with the plasma membrane, providing a mechanism to block initiation of pheromone response in cells already committed to entering the cell cycle [359, 360]. Mutations in a Ran binding protein (Yrb1) that preferentially prevent nuclear import and elevate the amount of Ste5 in the cytoplasm increase mating efficiency, suggesting that re-import of Ste5 back into the nucleus may also contribute to terminating signaling [361].

Fus3 purportedly binds to Gpa1, and mutations in Gpa1 that supposedly have no other effect on Gpa1 function other than to disrupt this interaction reportedly cause defects both in pheromone-induced polarized growth and in recovery from the stimulus [328]. On this basis, it was suggested that the Gα-Fus3 interaction may negatively regulate the activity of the MAPK, or may serve to channel its activity toward substrates at the plasma membrane (as opposed to those in the nucleus), or both [328]. Similarly, it has been reported [17] that Gpa1 interacts with Scp160, an mRNA-binding polysome-associated protein containing seven hnRNP K-homology (KH) repeats that localizes to the endoplasmic reticulum (ER) [362] and is an apparent ortholog of a mammalian RNP protein involved in nuclear RNA export, vigilin [363]. It is possible that this interaction might contribute to the differential recruitment of mRNAs to translating ribosomes that is observed during pheromone response, which was discussed above (see Section 4.5). Even more recently, it has been reported that Gpa1 released from pheromone receptors travels to endosomes and there encounters, binds to, and stimulates the PtdIns 3-kinase Vps34 and its regulator, Vps15 [21]. However, the resulting effects on cellular PtdIns3P levels are modest, even in cells expressing a GTP hydrolysis-defective Gpa1 mutant. Moreover, *vps34Δ* cells still mate at about ~25% the efficiency of their wild-type counterparts and exhibit only a modestly diminished capacity to mount a transcriptional response to pheromone, as judged from reporter gene assays. These effects are truly small compared to the phenotype of mutants defective in previously known components of the mating pathway. For example, a *ste4*, *ste5*, *ste7*, *ste11* or *ste12* mutant mates with a frequency that is at least seven orders of magnitude lower than that of wild-type cells [364].

Furthermore, Gpa1 is not required for the pheromone response pathway. In fact, *gpa1Δ* cells are unable to grow because they undergo G1 arrest and display other responses (shmoo formation) diagnostic of constitutive pathway response [365, 366], and all of these behaviors are eliminated in a *gpa1Δ ste4Δ* double mutant [367]. In other words, unlike Gα, Gβγ has an absolutely essential role in triggering all of the downstream events of pheromone response. Thus, although these newly described interactions of Gpa1 discussed in the preceding paragraph (with Fus3, with Scp160, and with Vps34) may contribute some sort of back-up mechanisms that enhance the robustness of the system overall, it is certainly the case that none of them is required for pheromone response, including the roles that Fus3 plays in projection formation.

5.4 Signal specificity and cross-pathway interactions

Different signal transduction pathways often utilize the same molecular components. For example,

Sho1, Msb2, Cdc24, Cdc42, Bem1, Ste20, Ste50, Ste11, Ste7, Kss1, and Ste12 are all utilized in at least two MAPK-mediated signal response pathways ([Fig. 1](#)). Accordingly, much recent research has been focused on elucidating the molecular bases underlying the distinction between and implementation of inter-pathway coordination and pathway specificity; recent reviews on this subject include [[368-371](#)].

In the larger and highly differentiated cells of metazoans, and in those cases wherein multiple pathways use a shared component, signal fidelity can be maintained by separating alternative downstream targets. This separation can be achieved, for example, by development-specific gene expression, such that a target is present in one cell type, but not in another; or, targets in the same cell can be separated spatially (for example, located at the apical as opposed to the baso-lateral surface) or temporally (for example, at different points in the cell cycle) (for further discussion, see [[369](#)]). The situation in yeast poses a more difficult situation with regard to establishing and maintaining specificity because the potentially confounding factors are all expressed together and are, relatively speaking, readily accessible to each other, as demonstrated by the fact that rather straightforward genetic mutations can give rise to abnormal cross-talk between pathways [[43](#), [372](#)]. So, under such especially “risky” circumstances, what mechanisms can help impose signaling fidelity?

5.4.1 Protein-protein interaction specificity.

One of the simplest mechanisms for regulating the choice of targets is differential binding affinity, or specificity of physical interactions. Enzyme-substrate or regulator-effector relationships often involve stable interactions between docking motifs, and otherwise similar proteins can be discriminated by relatively subtle differences in these motifs [[371](#), [373](#), [374](#)]. For example, during pheromone response, both Fus3 and Kss1 are activated (although the reason for this is still not entirely clear). Between the two, Fus3 is the primary mediator of the pheromone-induced cell cycle arrest, in part because the Cdk inhibitor Far1 is much more efficiently phosphorylated by Fus3 than by Kss1 [[46](#)]. This selectivity is not due to differences in the specific activities or catalytic efficiencies of Fus3 and Kss1 because the activated versions of both enzymes display equivalent specific activities on non-natural substrates, such as myelin basic protein [[46](#)]. Instead, as mentioned earlier, Far1 has a docking site that can recruit Fus3, but not Kss1 [[48](#)]. This simple mechanism allows the cell to discriminate the actions of quite similar enzymes, thereby imposing Fus3-dependent cell cycle arrest upon pheromone stimulation, but not allowing an inappropriate Kss1-induced arrest to occur during nutrient depletion when the cells need to continue to grow

invasively to be able to successfully forage for more nutrients.

The use of binding specificity to distinguish among similar proteins is highly prevalent among activator-scaffold and scaffold-kinase interactions. At the top of the HOG pathway, for example, the SH3 domain of the osmosensor Sho1 binds to a proline-rich motif on the scaffold Pbs2, and mutations that increase the strength of this interaction increase HOG pathway signaling [375]. Although this Pbs2 motif binds to SH3 domains from other organisms, the only yeast SH3 domain it binds to is that of Sho1, suggesting that specificity is a crucial feature of this interaction [376]. Consistent with this view, decreases in Sho1-Pbs2 affinity correlate with increases in the level of osmostress-induced cross-talk to pheromone pathway outputs [375]. Indeed, whereas mutations could be identified that increased the affinity of these partners, these mutants exhibited an increase in cross-reactivity, indicating that in the native situation, minimizing cross-activation may be more important than maximizing signal strength [376, 377].

At the level of MAPKKKs, Ste11 activates different MAPKKs in the pheromone response and HOG pathways. The choice of downstream pathway is restricted by the scaffold that Ste11 binds to in each case— when Ste11 is bound to the pheromone response scaffold, Ste5, signaling occurs only through the pheromone response pathway, but when Ste11 is associated with the HOG scaffold, Pbs2, signaling only occurs in the HOG pathway [378]. Consistent with this notion, artificially engineered scaffolds can redirect signal responses to unnatural outputs [379, 380]. At the level of MAPKs, although Fus3 and Slt2/Mpk1 are both homologs of the Erk subfamily of MAPKs, only Fus3 is activated by pheromone and only Slt2/Mpk1 by cell wall stress. This seems to be regulated at least in part by differential scaffold binding, as the CWI pathway scaffold Spa2 binds Slt2/Mpk1, but not Fus3, and the pheromone response pathway scaffold Ste5 binds Fus3, but not Slt2/Mpk1 [297]. Furthermore, such specificity is also manifest at the level of the MAPKKs. For example, Ste7 binds Fus3 (and Kss1) with apparently nanomolar affinity via its N-terminal MAPK docking site, whereas Ste7 does not interact detectably with Slt2/Mpk1 [381, 382].

The role of scaffold proteins is, however, not merely restrictive in the sense that they prevent activation of other pathways. In the pheromone response pathway, Fus3 activation requires the presence of Ste5, and overexpression of Ste5 channels inputs, either from constitutively active Ste11 or Ste11 activated by the HOG pathway under conditions (*pbs2Δ* cells) that permit cross-talk, even more efficiently to Fus3 [39]. Consistent with the requirement of Ste5 for Fus3 activation, in the absence of pheromone, expression of constitutively-active Ste7 variants leads to phosphorylation of Kss1, but not Fus3 [40]. This result has been attributed to the stronger affinity of certain states of Ste7 for Kss1 than for Fus3 [40] (for more on this point, see [381]), thus

rendering Fus3 activation dependent on an adaptor (Ste5) that binds both Fus3 and Ste7, whereas Kss1 does not need to bind to Ste5 to become activated, as also suggested by the findings of two other groups [41, 383]. The observation that a Ste5 fragment promotes partial Fus3 autoactivation [310] also raises the intriguing possibility that Fus3 and Kss1 may differ in their dependence on an allosteric activator. The ability of Kss1, but not Fus3, to be activated off of Ste5 (or any other known scaffold protein) is one basis for the avoidance of inappropriate Fus3-mediated responses during nutrient-depletion-stimulated filamentous growth.

A potential role for Ste50 in the maintenance of MAPK signal fidelity has been identified, although the mechanistic basis is unclear. For example, certain mutations in the SAM domain of Ste50, which binds to a SAM domain in Ste11, as discussed earlier, lead to decreased osmoresistance and increased cross-talk with the pheromone response and filamentous growth pathways upon hyperosmotic stress [384]. Casein kinase I phosphorylation of a specific threonine residue within the Ste50 SAM domain is required for pheromone response, but not for osmoresistance [385], further implicating differential Ste50 interactions in diverting Ste11 activation into the appropriate MAPK pathway.

5.4.2 Down-regulation of alternate pathways

Another mechanism by which an activated pathway can prevent activation of a similar pathway is to actively down-regulate the other route and thereby eliminate that alternative option altogether. Upon pheromone stimulation, Fus3 is activated for the duration of the cellular response, whereas Kss1 is activated transiently [43]. Although Kss1 activation contributes to downstream effects [43], Fus3 is thought to be the primary MAPK mediating pheromone-induced behaviors, as all available evidence suggests. Interestingly, the degree of Fus3 activity limits the magnitude and duration of Kss1 activation [43]. Given that activated Kss1 is required for filamentous growth, this situation has led to the hypothesis that transient Kss1 activity contributes to pheromone responses, but only prolonged Kss1 activity can support filamentous growth, and Fus3 actively regulates the duration that Kss1 is allowed to be active as a part of the normal pheromone response program [43]. The mechanism by which active Fus3 contributes to Kss1 inactivation is unknown, but the most likely candidate is activation of synthesis or function of a phosphatase that acts differentially on these two MAPKs. In another set of MAPK pathways, prevention of cross-talk between the osmostress and pheromone response pathways [372] requires the kinase activity of Hog1 [325], but not its nuclear translocation [101]. The relevant target(s) have not yet been identified, however.

In addition to inactivation, alternate pathways can be down-regulated by the degradation of their components. Upon pheromone stimulation, Fus3 phosphorylates Tec1, the critical transcription factor necessary for filamentous growth (but not for mating) ([Fig. 4](#)). This modification marks Tec1 for ubiquitin-mediated degradation [[176](#), [177](#), [386](#)]. In this way, there is no possibility for a cell responding to pheromone to engage in a filamentous growth response. As predicted by this conclusion, mutation of the specific residue in Tec1 that is phosphorylated by Fus3 to Ala stabilizes Tec1 in pheromone-treated cells and causes pheromone to elicit filamentous growth responses. It has also been reported that the stability of Tec1 is enhanced by SUMOylation [[387](#)], which is sometimes considered a modification that blocks ubiquitination [[388](#)]. Consistent with that view and the physiological sense of eliminating Tec1 in cells that are committed to a pheromone response, SUMOylation of Tec1 seems to be decreased upon pheromone stimulation [[387](#)]. Together, these mechanisms ensure that any pheromone-activated Kss1, which could otherwise potentially mediate filamentous growth, only has the option of activating the transcription of pheromone response genes. Compared to inactivation, degradation of factors that could mediate alternative responses is a more permanent mechanism of ensuring the specificity of a developmental behavior and is an elegant molecular model for the processes of commitment and cell fate determination that occur during the development of multicellular organisms.

5.4.3 MAPK pathways activating MAPK pathways

An instance of simultaneous regulation by and regulation of MAPKs occurs when one MAPK pathway activates another MAPK pathway as part of a sequential or composite signal response. Upon hyperosmotic shock and activation of Hog1, the Slt2/Mpk1 MAPK becomes activated, which appears to require the transcription factor, Rlm1 [[343](#)]. Similarly, upon pheromone stimulation and activation of Fus3, Slt2/Mpk1 becomes activated in a manner dependent on transcription and translation [[389](#)]. Interestingly, although pheromone-induced Slt2/Mpk1 activation requires Pkc1 and the redundant MAPKKs Mkk1 and Mkk2, it does not absolutely require the MAPKKK Bck1, suggesting that an alternative MAPKKK (most likely Ste11, given the circumstances) can serve to phosphorylate Mkk1 and Mkk2 [[389](#)]. It is thought that the secondary enlistment of the Slt2/Mpk1 pathway by the primary MAPK pathways during both the pheromone and osmotic stress response facilitates the cell wall alterations that must occur during the mating process and to cope with any damage incurred during the loss in turgor pressure under hyperosmotic conditions. It is not precisely known in either case how the primary pathway feeds in to activate the secondary pathway and at what level. Although the activation of one MAPK pathway by another is perhaps conceptually easier to understand than the antagonistic mechanisms that must occur between pathways to maintain signal fidelity, it highlights the fact that MAPK pathways are not always

competing, and often are coordinated in a positive manner.

6. Conclusions and perspectives

The MAPK pathways of yeast are among the best understood signal transduction pathways in biology. Recent studies have greatly advanced our knowledge about the functions and regulation of these pathways, particularly in the areas of cell cycle regulation, transcriptional regulation, translational regulation, regulation by MAPK phosphatases, and maintenance of signal specificity.

However, a great many questions remain. What are the direct activators of Smk1— does a classical MAPK cascade, or a scaffold, exist in this pathway? How is Cdc42 activated in the filamentous growth and HOG pathways? What are the direct ligands or physical stresses that activate the filamentous growth, HOG, and CWI pathways, and how do their sensors mechanistically transduce these extracellular or membrane-based signals to their intracellular interaction partners?

What are the composition and arrangement of scaffolded MAPK complexes during pathway activation and inactivation? Is allosteric activation by scaffold proteins of their bound kinases important in native situations, and do scaffold proteins have additional active roles in signal transduction?

What unique roles, if any, does Kss1 perform during pheromone response in wild-type cells? Are there additional molecular coordination between the MAPK, PKA, and AMPK pathways in the regulation of filamentous growth at a level other than transcription?

Does the slight increase in Fus3 nuclear localization upon pheromone stimulation contribute anything to pathway activation? What determines the subcellular distribution of MAPKs that seem to be regulated independently of classical Ran-karyopherin mechanisms?

What phosphatases dephosphorylate Kss1, and how does Fus3 limit Kss1 activation during pheromone response? What does active Hog1 target to prevent cross-talk between the HOG and pheromone response pathways? More generally, what mechanisms exist, if any, by which MAPKs actively inhibit activation of other MAPKs to maintain signal fidelity?

What different mechanisms are employed by MAPKs in post-transcriptional and translational

regulation of gene expression, and are how important is regulation at these levels compared to regulation of transcription initiation?

Many, if not most, MAPK targets are probably still unidentified. In particular, given the cellular processes that are known to occur upon pathway stimulation and are dependent upon activation of MAPKs, we are likely still missing as direct MAPK substrates many metabolic enzymes, cytoskeletal and cell surface proteins, and gene expression factors beyond the class of regulators of transcription initiation.

As researchers continue to address these and other outstanding questions, and continue to identify and characterize new targets and regulatory mechanisms of the MAPK pathways of yeast, we are confident that studies in this model system will continue to play a leading role in establishing our understanding of the molecular basis of cellular signal responses in all organisms.

Acknowledgements

Work from the corresponding author's laboratory on MAPK signaling in yeast has been supported by NIH Predoctoral Traineeship GM07232 (to R.E.C.) and by NIH Research Grant GM21841 (to J.T.). We thank members of the Thorner laboratory, especially Lindsay Garrenton and Michael McMurray, for helpful discussions. We apologize in advance to any colleagues in the field whose contributions may have been overlooked or whose findings we may have interpreted differently.

Footnotes

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final citable form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

References

1. Widmann C, Gibson S, Jarpe MB, Johnson GL. *Physiol Rev.* 1999;79:143–80. doi:

- 10.1152/physrev.1999.79.1.143. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Lamson RE, Winters MJ, Pryciak PM. Mol Cell Biol. 2002;22:2939–51. doi: 10.1128/MCB.22.9.2939-2951.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Wu C, Whiteway M, Thomas DY, Leberer E. J Biol Chem. 1995;270:15984–92. doi: 10.1074/jbc.270.27.15984. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
4. Drogen F, O'Rourke SM, Stucke VM, Jaquenoud M, Neiman AM, Peter M. Curr Biol. 2000;10:630–9. doi: 10.1016/s0960-9822(00)00511-x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
5. Winters MJ, Pryciak PM. Mol Cell Biol. 2005;25:2177–90. doi: 10.1128/MCB.25.6.2177-2190.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Leeuw T, Wu C, Schrag JD, Whiteway M, Thomas DY, Leberer E. Nature. 1998;391:191–5. doi: 10.1038/34448. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Wu C, Leberer E, Thomas DY, Whiteway M. Mol Biol Cell. 1999;10:2425–40. doi: 10.1091/mbc.10.7.2425. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Truckses DM, Bloomekatz JE, Thorner J. Mol Cell Biol. 2006;26:912–28. doi: 10.1128/MCB.26.3.912-928.2006. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Tatebayashi K, Yamamoto K, Tanaka K, Tomida T, Maruoka T, Kasukawa E, Saito H. Embo J. 2006;25:3033–44. doi: 10.1038/sj.emboj.7601192. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Choi KY, Satterberg B, Lyons DM, Elion EA. Cell. 1994;78:499–512. doi: 10.1016/0092-8674(94)90427-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
11. Marcus S, Polverino A, Barr M, Wigler M. Proc Natl Acad Sci U S A. 1994;91:7762–6. doi: 10.1073/pnas.91.16.7762. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
12. Printen JA, Sprague GF, Jr. Genetics. 1994;138:609–19. doi: 10.1093/genetics/138.3.609. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

13. Whiteway MS, Wu C, Leeuw T, Clark K, Fourest-Lieuvin A, Thomas DY, Leberer E. *Science*. 1995;269:1572–5. doi: 10.1126/science.7667635. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
14. Zarrinpar A, Bhattacharyya RP, Nittler MP, Lim WA. *Mol Cell*. 2004;14:825–32. doi: 10.1016/j.molcel.2004.06.011. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Qi M, Elion EA. *J Cell Sci*. 2005;118:3569–72. doi: 10.1242/jcs.02470. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
16. Hoffman CS. *Eukaryot Cell*. 2005;4:495–503. doi: 10.1128/EC.4.3.495-503.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
17. Guo M, Aston C, Burchett SA, Dyke C, Fields S, Rajarao SJ, Uetz P, Wang Y, Young K, Dohlman HG. *Mol Cell*. 2003;12:517–24. doi: 10.1016/s1097-2765(03)00307-1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
18. Bar EE, Ellicott AT, Stone DE. *J Biol Chem*. 2003;278:21798–804. doi: 10.1074/jbc.M212636200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
19. Butty AC, Pryciak PM, Huang LS, Herskowitz I, Peter M. *Science*. 1998;282:1511–6. doi: 10.1126/science.282.5393.1511. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
20. Nern A, Arkowitz RA. *J Cell Biol*. 1999;144:1187–202. doi: 10.1083/jcb.144.6.1187. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
21. Slessareva JE, Routt SM, Temple B, Bankaitis VA, Dohlman HG. *Cell*. 2006;126:191–203. doi: 10.1016/j.cell.2006.04.045. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Bardwell L. *Peptides*. 2005;26:339–50. doi: 10.1016/j.peptides.2004.10.002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
23. Wang Y, Dohlman HG. *Science*. 2004;306:1508–9. doi: 10.1126/science.1104568. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
24. Dohlman HG, Thorner JW. *Annu Rev Biochem*. 2001;70:703–54. doi: 10.1146/annurev.biochem.70.1.703. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Elion EA. *Curr Opin Microbiol*. 2000;3:573–81. doi: 10.1016/s1369-5274(00)00143-

0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

26. Dohlman HG. *Annu Rev Physiol.* 2002;64:129–52. doi:

10.1146/annurev.physiol.64.081701.133448. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

27. Klein S, Reuveni H, Levitzki A. *Proc Natl Acad Sci U S A.* 2000;97:3219–23. doi:

10.1073/pnas.050015797. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

28. Inouye C, Dhillon N, Thorner J. *Science.* 1997;278:103–6. doi:

10.1126/science.278.5335.103. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

29. Feng Y, Song LY, Kincaid E, Mahanty SK, Elion EA. *Curr Biol.* 1998;8:267–78. doi:

10.1016/s0960-9822(98)70108-3. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

30. Wiget P, Shimada Y, Butty AC, Bi E, Peter M. *Embo J.* 2004;23:1063–74. doi:

10.1038/sj.emboj.7600123. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

31. van Drogen-Petit A, Zwahlen C, Peter M, Bonvin AM. *J Mol Biol.* 2004;336:1195–210.

doi: 10.1016/j.jmb.2003.12.062. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

32. Winters MJ, Lamson RE, Nakanishi H, Neiman AM, Pryciak PM. *Mol. Cell.*

2005;20:21–32. doi: 10.1016/j.molcel.2005.08.020. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

33. Garrenton LS, Young SL, Thorner J. *Genes Dev.* 2006;20:1946–58. doi:

10.1101/gad.1413706. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

34. Finegold AA, Schafer WR, Rine J, Whiteway M, Tamanai F. *Science.* 1990;249:165–9.

doi: 10.1126/science.1695391. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

35. Hirschman JE, Jenness DD. *Mol Cell Biol.* 1999;19:7705–11. doi:

10.1128/mcb.19.11.7705. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

36. Manahan CL, Patnana M, Blumer KJ, Linder ME. *Mol Biol Cell.* 2000;11:957–68. doi:

10.1091/mbc.11.3.957. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

37. Ma D, Cook JG, Thorner J. *Mol Biol Cell.* 1995;6:889–909. doi: 10.1091/mbc.6.7.889.

[[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

38. Gagiano M, Bauer FF, Pretorius IS. *FEMS Yeast Res.* 2002;2:433–70. doi:

- 10.1111/j.1567-1364.2002.tb00114.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
39. Flatauer LJ, Zadeh SF, Bardwell L. Mol Cell Biol. 2005;25:1793–803. doi: 10.1128/MCB.25.5.1793-1803.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
40. Maleri S, Ge Q, Hackett EA, Wang Y, Dohlman HG, Errede B. Mol Cell Biol. 2004;24:9221–38. doi: 10.1128/MCB.24.20.9221-9238.2004. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
41. Andersson J, Simpson DM, Qi M, Wang Y, Elion EA. Embo J. 2004;23:2564–76. doi: 10.1038/sj.emboj.7600250. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
42. Elion EA, Brill JA, Fink GR. Proc Natl Acad Sci U S A. 1991;88:9392–6. doi: 10.1073/pnas.88.21.9392. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
43. Sabbagh W, Jr., Flatauer LJ, Bardwell AJ, Bardwell L. Mol Cell. 2001;8:683–91. doi: 10.1016/s1097-2765(01)00322-7. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
44. Breitskreutz A, Boucher L, Breitskreutz BJ, Sultan M, Jurisica I, Tyers M. Genetics. 2003;165:997–1015. doi: 10.1093/genetics/165.3.997. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
45. Truckses DM, Garrenton LS, Thorner J. Science. 2004;306:1509–11. doi: 10.1126/science.1104677. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
46. Breitskreutz A, Boucher L, Tyers M. Curr Biol. 2001;11:1266–71. doi: 10.1016/s0960-9822(01)00370-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
47. Cherkasova V, Lyons DM, Elion EA. Genetics. 1999;151:989–1004. doi: 10.1093/genetics/151.3.989. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
48. Remenyi A, Good MC, Bhattacharyya RP, Lim WA. Mol Cell. 2005;20:951–62. doi: 10.1016/j.molcel.2005.10.030. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
49. Colman-Lerner A, Gordon A, Serra E, Chin T, Resnekov O, Endy D, Pesce CG, Brent R. Nature. 2005;437:699–706. doi: 10.1038/nature03998. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

50. Erdman S, Snyder M. *Genetics*. 2001;159:919–28. doi: 10.1093/genetics/159.3.919. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
51. Verstrepen KJ, Klis FM. *Mol. Microbiol.* 2006;60:5–15. doi: 10.1111/j.1365-2958.2006.05072.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
52. Schnepfer L, Duvel K, Broach JR. *Curr Opin Microbiol.* 2004;7:624–30. doi: 10.1016/j.mib.2004.10.002. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
53. Palecek SP, Parikh AS, Kron SJ. *Microbiology*. 2002;148:893–907. doi: 10.1099/00221287-148-4-893. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
54. Gancedo JM. *FEMS Microbiol Rev.* 2001;25:107–23. doi: 10.1111/j.1574-6976.2001.tb00573.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
55. Pan X, Harashima T, Heitman J. *Curr Opin Microbiol.* 2000;3:567–72. doi: 10.1016/s1369-5274(00)00142-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
56. Lengeler KB, Davidson RC, D'Souza C, Harashima T, Shen WC, Wang P, Pan X, Waugh M, Heitman J. *Microbiol Mol Biol Rev.* 2000;64:746–85. doi: 10.1128/mmbr.64.4.746-785.2000. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
57. Mosch HU, Roberts RL, Fink GR. *Proc Natl Acad Sci U S A.* 1996;93:5352–6. doi: 10.1073/pnas.93.11.5352. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
58. Cullen PJ, Sabbagh W, Jr., Graham E, Irick MM, van Olden EK, Neal C, Delrow J, Bardwell L, Sprague GF, Jr. *Genes Dev.* 2004;18:1695–708. doi: 10.1101/gad.1178604. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
59. Lorenz MC, Heitman J. *Embo J.* 1998;17:1236–47. doi: 10.1093/emboj/17.5.1236. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
60. Van Nuland A, Vandormael P, Donaton M, Alenquer M, Lourenco A, Quintino E, Versele M, Thevelein JM. *Mol Microbiol.* 2006;59:1485–505. doi: 10.1111/j.1365-2958.2005.05043.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
61. Tamaki H, Miwa T, Shinozaki M, Saito M, Yun CW, Yamamoto K, Kumagai H. *Biochem Biophys Res Commun.* 2000;267:164–8. doi: 10.1006/bbrc.1999.1914. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

62. Lorenz MC, Pan X, Harashima T, Cardenas ME, Xue Y, Hirsch JP, Heitman J. *Genetics*. 2000;154:609–22. doi: 10.1093/genetics/154.2.609. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
63. Maeda T, Takekawa M, Saito H. *Science*. 1995;269:554–8. doi: 10.1126/science.7624781. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
64. Wu C, Jansen G, Zhang J, Thomas DY, Whiteway M. *Genes Dev*. 2006;20:734–46. doi: 10.1101/gad.1375706. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
65. O'Rourke SM, Herskowitz I. *Mol Cell Biol*. 2002;22:4739–49. doi: 10.1128/MCB.22.13.4739-4749.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
66. Wang X, Rickert M, Garcia KC. *Science*. 2005;310:1159–63. doi: 10.1126/science.1117893. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
67. Soupene E, Ramirez RM, Kustu S. *Mol Cell Biol*. 2001;21:5733–41. doi: 10.1128/MCB.21.17.5733-5741.2001. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
68. Lemaire K, Van de Velde S, Van Dijck P, Thevelein JM. *Mol Cell*. 2004;16:293–9. doi: 10.1016/j.molcel.2004.10.004. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
69. Xue Y, Batlle M, Hirsch JP, Embo J. 1998;17:1996–2007. doi: 10.1093/emboj/17.7.1996. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
70. Harashima T, Heitman J. *Mol Biol Cell*. 2005;16:4557–71. doi: 10.1091/mbc.E05-05-0403. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
71. Harashima T, Heitman J. *Mol Cell*. 2002;10:163–73. doi: 10.1016/s1097-2765(02)00569-5. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
72. Toda T, Uno I, Ishikawa T, Powers S, Kataoka T, Broek D, Cameron S, Broach J, Matsumoto K, Wigler M. *Cell*. 1985;40:27–36. doi: 10.1016/0092-8674(85)90305-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
73. Harashima T, Anderson S, Yates JR, 3rd, Heitman J. *Mol Cell*. 2006;22:819–30. doi: 10.1016/j.molcel.2006.05.011. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

74. Lu A, Hirsch JP. *Eukaryot Cell*. 2005;4:1794–800. doi: 10.1128/EC.4.11.1794-1800.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
75. Peeters T, Louwet W, Gelade R, Nauwelaers D, Thevelein JM, Versele M. *Proc Natl Acad Sci U S A*. 2006;103:13034–9. doi: 10.1073/pnas.0509644103. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
76. Dickinson JR. *Microbiology*. 1996;142(Pt 6):1391–7. doi: 10.1099/13500872-142-6-1391. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
77. Lorenz MC, Cutler NS, Heitman J. *Mol Biol Cell*. 2000;11:183–99. doi: 10.1091/mbc.11.1.183. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
78. Chen H, Fink GR. *Genes Dev*. 2006;20:1150–61. doi: 10.1101/gad.1411806. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
79. Hardie DG, Carling D, Carlson M. *Annu Rev Biochem*. 1998;67:821–55. doi: 10.1146/annurev.biochem.67.1.821. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
80. Kuchin S, Vyas VK, Carlson M. *Mol Cell Biol*. 2002;22:3994–4000. doi: 10.1128/MCB.22.12.3994-4000.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
81. Vyas VK, Kuchin S, Berkey CD, Carlson M. *Mol Cell Biol*. 2003;23:1341–8. doi: 10.1128/MCB.23.4.1341-1348.2003. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
82. Bayly JC, Douglas LM, Pretorius IS, Bauer FF, Dranginis AM. *FEMS Yeast Res*. 2005;5:1151–6. doi: 10.1016/j.femsyr.2005.05.004. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
83. Orlova M, Kanter E, Krakovich D, Kuchin S. *Eukaryot Cell*. 2006;5:1831–7. doi: 10.1128/EC.00110-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
84. Han J, Lee JD, Bibbs L, Ulevitch RJ. *Science*. 1994;265:808–11. doi: 10.1126/science.7914033. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
85. Westfall PJ, Ballon DR, Thorner J. *Science*. 2004;306:1511–2. doi: 10.1126/science.1104879. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

86. Saito H, Tatebayashi K. *J Biochem (Tokyo)* 2004;136:267–72. doi: 10.1093/jb/mvh135. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
87. O'Rourke SM, Herskowitz I, O'Shea EK. *Trends Genet.* 2002;18:405–12. doi: 10.1016/s0168-9525(02)02723-3. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
88. Hohmann S. *Microbiol. Mol. Biol. Rev.* 2002;66:300–372. doi: 10.1128/MMBR.66.2.300-372.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
89. de Nadal E, Alepuz PM, Posas F. *EMBO Rep.* 2002;3:735–40. doi: 10.1093/embo-reports/kvf158. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
90. Posas F, Wurgler-Murphy SM, Maeda T, Witten EA, Thai TC, Saito H. *Cell.* 1996;86:865–75. doi: 10.1016/s0092-8674(00)80162-2. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
91. Posas F, Saito H. *Embo J.* 1998;17:1385–94. doi: 10.1093/emboj/17.5.1385. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
92. O'Rourke SM, Herskowitz I. *Mol. Biol. Cell.* 2004;15:532–542. doi: 10.1091/mbc.E03-07-0521. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
93. Posas F, Saito H. *Science.* 1997;276:1702–5. doi: 10.1126/science.276.5319.1702. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
94. Kwan JJ, Warner N, Maini J, Chan Tung KW, Zakaria H, Pawson T, Donaldson LW. *J Mol Biol.* 2006;356:142–54. doi: 10.1016/j.jmb.2005.11.012. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
95. Tatebayashi K, Takekawa M, Saito H. *EMBO J.* 2003;22:3624–3634. doi: 10.1093/emboj/cdg353. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
96. Bhattacharyya RP, Remenyi A, Yeh BJ, Lim WA. *Annu. Rev. Biochem.* 2006;75:655–680. doi: 10.1146/annurev.biochem.75.103004.142710. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
97. Chang CI, Xu BE, Akella R, Cobb MH, Goldsmith EJ. *Mol. Cell.* 2002;9:1241–1249. doi: 10.1016/s1097-2765(02)00525-7. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

98. Raitt DC, Posas F, Saito H, Embo J. 2000;19:4623–31. doi: 10.1093/emboj/19.17.4623. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
99. Reiser V, Salah SM, Ammerer G. Nat Cell Biol. 2000;2:620–7. doi: 10.1038/35023568. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
100. Ferrigno P, Posas F, Koepp D, Saito H, Silver PA. Embo J. 1998;17:5606–14. doi: 10.1093/emboj/17.19.5606. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
101. Westfall PJ, Patterson JC, Chen RE, Thorner J. Nature Cell Biol. 2007 submitted. [[Google Scholar](#)]
102. Bilsland-Marchesan E, Arino J, Saito H, Sunnerhagen P, Posas F. Mol Cell Biol. 2000;20:3887–95. doi: 10.1128/mcb.20.11.3887-3895.2000. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
103. Teige M, Scheikl E, Reiser V, Ruis H, Ammerer G. Proc Natl Acad Sci U S A. 2001;98:5625–30. doi: 10.1073/pnas.091610798. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
104. de Nadal E, Casadome L, Posas F. Mol Cell Biol. 2003;23:229–37. doi: 10.1128/MCB.23.1.229-237.2003. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
105. Harrison JC, Zyla TR, Bardes ES, Lew DJ. J Biol Chem. 2004;279:2616–22. doi: 10.1074/jbc.M306110200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
106. Levin DE. Microbiol Mol Biol Rev. 2005;69:262–91. doi: 10.1128/MMBR.69.2.262-291.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
107. Garcia R, Bermejo C, Grau C, Perez R, Rodriguez-Pena JM, Francois J, Nombela C, Arroyo J. J Biol Chem. 2004;279:15183–95. doi: 10.1074/jbc.M312954200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
108. Lesage G, Bussey H. Microbiol. Mol. Biol. Rev. 2006;70:317–343. doi: 10.1128/MMBR.00038-05. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
109. Torres L, Martin H, Garcia-Saez MI, Arroyo J, Molina M, Sanchez M, Nombela C. Mol Microbiol. 1991;5:2845–54. doi: 10.1111/j.1365-2958.1991.tb01993.x. [[DOI](#)]

[[PubMed](#)] [[Google Scholar](#)]

110. Philip B, Levin DE. *Mol Cell Biol*. 2001;21:271–80. doi: 10.1128/MCB.21.1.271-280.2001. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

111. Ozaki K, Tanaka K, Imamura H, Hihara T, Kameyama T, Nonaka H, Hirano H, Matsuura Y, Takai Y, Embo J. 1996;15:2196–207. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

112. Heo WD, Inoue T, Park WS, Kim ML, Park BO, Wandless TJ, Meyer T. *Science*. 2006;314:1458–1461. doi: 10.1126/science.1134389. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

113. Richman TJ, Sawyer MM, Johnson DI. *Eukaryot Cell*. 2002;1:458–68. doi: 10.1128/EC.1.3.458-468.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

114. Ubersax JA, Woodbury EL, Quang PN, Paraz M, Blethrow JD, Shah K, Shokat KM, Morgan DO. *Nature*. 2003;425:859–864. doi: 10.1038/nature02062. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

115. Yoshida S, Kono K, Lowery DM, Bartolini S, Yaffe MB, Ohya Y, Pellman D. *Science*. 2006;313:108–111. doi: 10.1126/science.1126747. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

116. Lorberg A, Schmitz HP, Gengenbacher U, Heinisch JJ. *Yeast*. 2003;20:611–24. doi: 10.1002/yea.989. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

117. Audhya A, Emr SD. *Dev. Cell*. 2002;2:593–605. doi: 10.1016/s1534-5807(02)00168-5. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

118. Lorberg A, Jacoby JJ, Schmitz HP, Heinisch JJ. *Mol Genet Genomics*. 2001;266:505–13. doi: 10.1007/s004380100579. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

119. Vay HA, Philip B, Levin DE. *Microbiology*. 2004;150:3281–3288. doi: 10.1099/mic.0.27264-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

120. Schmelzle T, Helliwell SB, Hall MN. *Mol Cell Biol*. 2002;22:1329–39. doi: 10.1128/mcb.22.5.1329-1339.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

121. Roelants FM, Torrance PD, Bezman N, Thorner J. *Mol Biol Cell*. 2002;13:3005–28. doi: 10.1091/mbc.E02-04-0201. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
122. Matsuo T, Kubo Y, Watanabe Y, Y. M. *EMBO J*. 2003;22:3073–3083. doi: 10.1093/emboj/cdg298. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
123. Loewith R, Jacinto E, Wullschlegel S, Lorberg A, Crespo JL, Bonenfant D, Oppliger W, Jenoe P, Hall MN. *Mol. Cell*. 2002;10:457–468. doi: 10.1016/s1097-2765(02)00636-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
124. Audhya A, Loewith R, Parsons AB, Gao L, Tabuchi M, Zhou H, Boone C, Hall MN, Emr SD. *Embo J*. 2004;23:3747–57. doi: 10.1038/sj.emboj.7600384. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
125. Fadri M, Daquinag A, Wang S, Xue T, Kunz J. *Mol Biol Cell*. 2005;16:1883–900. doi: 10.1091/mbc.E04-07-0564. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
126. deHart AK, Schnell JD, Allen DA, Tsai JY, Hicke L. *Mol. Biol. Cell*. 2003;14:4676–4684. doi: 10.1091/mbc.E03-05-0323. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
127. Ho HL, Shiau YS, Chen MY. *Curr. Genet*. 2005;47:273–288. doi: 10.1007/s00294-005-0570-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
128. Jacinto E, Loewith R, Schmidt A, Lin S, Ruegg MA, Hall A, Hall MN. *Nature Cell Biol*. 2004;6:1122–1128. doi: 10.1038/ncb1183. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
129. Roelants FM, Torrance PD, Thorner J. *Microbiology*. 2004;150:3289–304. doi: 10.1099/mic.0.27286-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
130. Bultynck G, Heath VL, Majeed AP, Galan JM, Haguenaue-Tsapis R, Cyert MS. *Mol. Cell. Biol*. 2006;26:4729–4745. doi: 10.1128/MCB.01973-05. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
131. Daquinag A, Fadri M, Jung SY, Q. J., Kunz J. *Mol. Cell. Biol*. 2007;27:633–650. doi: 10.1128/MCB.00461-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
132. Parker PJ, Murray-Rust J. *J. Cell Sci*. 2004;117:131–132. doi: 10.1242/jcs.00982.

[[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

133. Vincent S, Settleman J. *Mol. Cell. Biol.* 1997;2247–2256. doi:

10.1128/mcb.17.4.2247. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

134. Mellor H, Parker PJ. *Biochem. J.* 1998;332:281–292. doi: 10.1042/bj3320281.

[[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

135. Kamada Y, Qadota H, Python C, Anraku Y, Ohya Y, Levin DE. *J. Biol. Chem.*

1996;271:9193–9196. doi: 10.1074/jbc.271.16.9193. [[DOI](#)] [[PubMed](#)] [[Google](#)

[Scholar](#)]

136. Fields FO. *Biochemistry*. University of California; Berkeley, CA: 1991. p. 257. Vol.

Ph.D. [[Google Scholar](#)]

137. Errede B, Cade RM, Yashar BM, Kamada Y, Levin DE, Irie K, Matsumoto K. *Mol*

Reprod Dev. 1995;42:477–85. doi: 10.1002/mrd.1080420416. [[DOI](#)] [[PubMed](#)]

[[Google Scholar](#)]

138. Martin H, Rodriguez-Pachon JM, Ruiz C, Nombela C, Molina M. *J Biol Chem.*

2000;275:1511–9. doi: 10.1074/jbc.275.2.1511. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

139. Jung US, Levin DE. *Mol Microbiol.* 1999;34:1049–57. doi: 10.1046/j.1365-

2958.1999.01667.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

140. Dodou E, Treisman R. *Mol Cell Biol.* 1997;17:1848–59. doi: 10.1128/mcb.17.4.1848.

[[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

141. Jung US, Sobering AK, Romeo MJ, Levin DE. *Mol Microbiol.* 2002;46:781–9. doi:

10.1046/j.1365-2958.2002.03198.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

142. Madden K, Sheu YJ, Baetz K, Andrews B, Snyder M. *Science.* 1997;275:1781–4. doi:

10.1126/science.275.5307.1781. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

143. Cyert MS, Kunisawa R, Kaim D, Thorner J. *Proc Natl Acad Sci U S A.* 1991;88:7376–

80. doi: 10.1073/pnas.88.16.7376. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google](#)

[Scholar](#)]

144. Cyert MS, Thorner J. *Mol Cell Biol.* 1992;12:3460–9. doi: 10.1128/mcb.12.8.3460.

[[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

145. Cyert MS. Biochem. Biophys. Res. Commun. 2003;311:1143–1150. doi: 10.1016/s0006-291x(03)01552-3. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

146. Zhao C, Jung US, Garrett-Engle P, Roe T, Cyert MS, Levin DE. Mol. Cell. Biol. 1998;18:1013–1022. doi: 10.1128/mcb.18.2.1013. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

147. Bonilla M, Cunningham KW. Mol Biol Cell. 2003;14:4296–305. doi: 10.1091/mbc.E03-02-0113. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

148. Garcia-Rodriguez LJ, Valle R, Duran A, Roncero C. FEBS Letts. 2005;579:6186–6190. doi: 10.1016/j.febslet.2005.10.001. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

149. Claret S, Gatti X, Doignon F, Thoraval D, Crouzet M. Eukaryot. Cell. 2005;4:1375–86. doi: 10.1128/EC.4.8.1375-1386.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

150. Bermejo C, Rodriguez E, Garcia R, Arias PS, A.B., Blanco N, Diez S, Rodriguez-Pena JM, Nombela C, Arroyo J. Genetics Society of America. Princeton University; Princeton, NJ: 2006. Yeast Genetics and Molecular Biology Meeting; p. 53. [[Google Scholar](#)]

151. Krasley E, Cooper KF, Mallory MJ, Dunbrack R, Strich R. Genetics. 2006;172:1477–86. doi: 10.1534/genetics.105.052266. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

152. Millson SH, Truman AW, King V, Prodromou C, Pearl LH, Piper PW. Eukaryot Cell. 2005;4:849–60. doi: 10.1128/EC.4.5.849-860.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

153. Truman AW, Millson SH, Nuttall JM, Mollapour M, Prodromou C, Piper PW. Eukaryot. Cell. 2007 doi: 10.1128/EC.00009-07. [Epub ahead of print, 9 Febr. 07] [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

154. Hawle P, Horst D, Bebelman JP, Yang XX, Siderius M, van der Vies SM. Eukaryot. Cell. 2007 doi: 10.1128/EC.00343-06. [Epub ahead of print, 12 Jan 07] [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

155. Lee P, Shabbir A, Cardozo C, Caplan AJ. *Mol Biol Cell*. 2004;15:1785–92. doi: 10.1091/mbc.E03-07-0480. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
156. Freese EB, Chu MI, Freese E. *J Bacteriol*. 1982;149:840–51. doi: 10.1128/jb.149.3.840-851.1982. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
157. Kupiec M, Byers B, Esposito RE, Mitchell AP. In: *The molecular and cellular biology of the yeast Saccharomyces*. Pringle JR, Broach JR, Jones EW, editors. Cold Spring Harbor Laboratory Press; Cold Spring Harbor: 1997. pp. 889–1936. [[Google Scholar](#)]
158. Neiman AM. *Microbiol. Mol. Biol. Rev.* 2005;69:565–584. doi: 10.1128/MMBR.69.4.565-584.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
159. Krisak L, Strich R, Winters RS, Hall JP, Mallory MJ, Kreitzer D, Tuan RS, Winter E. *Genes Dev*. 1994;8:2151–61. doi: 10.1101/gad.8.18.2151. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
160. Huang LS, Doherty HK, Herskowitz I. *Proc Natl Acad Sci U S A*. 2005;102:12431–6. doi: 10.1073/pnas.0502324102. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
161. Wagner M, Briza P, Pierce M, Winter E. *Genetics*. 1999;151:1327–40. doi: 10.1093/genetics/151.4.1327. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
162. Shaw JA, Mol PC, Bowers B, Silverman SJ, Valdivieso MH, Duran A, Cabib E. *J Cell Biol*. 1991;114:111–23. doi: 10.1083/jcb.114.1.111. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
163. Choi WJ, Santos B, Duran A, Cabib E. *Mol Cell Biol*. 1994;14:7685–94. doi: 10.1128/mcb.14.12.7685. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
164. Bungard D, Reed M, Winter E. *Eukaryot Cell*. 2004;3:910–8. doi: 10.1128/EC.3.4.910-918.2004. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
165. Pierce M, Wagner M, Xie J, Gailus-Durner V, Six J, Vershon AK, Winter E. *Mol Cell Biol*. 1998;18:5970–80. doi: 10.1128/mcb.18.10.5970. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

166. Schaber M, Lindgren A, Schindler K, Bungard D, Kaldis P, Winter E. *Mol Cell Biol.* 2002;22:57–68. doi: 10.1128/MCB.22.1.57-68.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
167. McDonald CM, Cooper KE, Winter E. *Genetics.* 2005;171:901–11. doi: 10.1534/genetics.105.045567. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
168. Friesen H, Lunz R, Doyle S, Segall J. *Genes Dev.* 1994;8:2162–75. doi: 10.1101/gad.8.18.2162. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
169. Gustin MC, Albertyn J, Alexander M, Davenport K. *Microbiol Mol Biol Rev.* 1998;62:1264–300. doi: 10.1128/mmbr.62.4.1264-1300.1998. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
170. Chu S, DeRisi J, Eisen M, Mulholland J, Botstein D, Brown PO, Herskowitz I. *Science.* 1998;282:699–705. doi: 10.1126/science.282.5389.699. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
171. Hepworth SR, Friesen H, Segall J. *Mol. Cell. Biol.* 1998;18:5750–5761. doi: 10.1128/mcb.18.10.5750. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
172. Iwamoto MA, Fairclough SR, Rudge SA, Engebrecht J. *Eukaryot Cell.* 2005;4:536–44. doi: 10.1128/EC.4.3.536-544.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
173. Proft M, Struhl K. *Cell.* 2004;118:351–61. doi: 10.1016/j.cell.2004.07.016. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
174. Kinclova-Zimmermannova O, Sychrova H. *Curr. Genet.* 2006;49:229–236. doi: 10.1007/s00294-005-0050-1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
175. Thorsen M, Di Y, Tangemo C, Morillas M, Ahmadpour D, Van der Does C, Wagner A, Johansson E, Boman J, Posas F, Wysocki R, Tamas MJ. *Mol. Biol. Cell.* 2006;17:4400–4410. doi: 10.1091/mbc.E06-04-0315. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
176. Bao MZ, Schwartz MA, Cantin GT, Yates JR, 3rd, Madhani HD. *Cell.* 2004;119:991–1000. doi: 10.1016/j.cell.2004.11.052. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

177. Chou S, Huang L, Liu H. *Cell*. 2004;119:981–90. doi: 10.1016/j.cell.2004.11.053. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
178. Friesen H, Murphy K, Breitskreutz A, Tyers M, Andrews B. *Mol Biol Cell*. 2003;14:3027–40. doi: 10.1091/mbc.E02-09-0613. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
179. Huang D, Patrick G, Moffat J, Tsai LH, Andrews B. *Proc. Natl. Acad. Sci. USA*. 1999;96:14445–14450. doi: 10.1073/pnas.96.25.14445. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
180. Lee J, Colwill K, Aneliunas V, Tennyson C, Moore L, Ho Y, Andrews B. *Curr Biol*. 1998;8:1310–21. doi: 10.1016/s0960-9822(07)00561-1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
181. Winter D, Lechler T, Li R. *Curr. Biol*. 1999;9:501–504. doi: 10.1016/s0960-9822(99)80218-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
182. Stefan CJ, Padilla SM, Audhya A, Emr SD. *Mol. Cell. Biol*. 2005;25:2910–2923. doi: 10.1128/MCB.25.8.2910-2923.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
183. Brizzio V, Gammie AE, Rose MD. *J Cell Biol*. 1998;141:567–84. doi: 10.1083/jcb.141.3.567. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
184. Heiman MG, Engel A, Walter P. *J. Cell Biol*. 2007;176:209–222. doi: 10.1083/jcb.200609182. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
185. Nolan S, Cowan AE, Koppel DE, Jin H, Grote E. *Mol. Biol. Cell*. 2006;17:2439–2450. doi: 10.1091/mbc.E05-11-1015. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
186. Lee MC, Schekman R. *Science*. 2004;303:479–480. doi: 10.1126/science.1094231. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
187. Crouzet M, Urdaci M, Dulau L, Aigle M. *Yeast*. 1991;7:727–43. doi: 10.1002/yea.320070708. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
188. Bauer F, Urdaci M, Aigle M, Crouzet M. *Mol Cell Biol*. 1993;13:5070–84. doi:

- 10.1128/mcb.13.8.5070. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
189. Bidlingmaier S, Snyder M. *J Cell Biol.* 2004;164:207–18. doi: 10.1083/jcb.200307065. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
190. Tcheperegine SE, Gao XD, Bi E. *Mol. Cell. Biol.* 2005;25:8567–8580. doi: 10.1128/MCB.25.19.8567-8580.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
191. Evangelista M, Blundell K, Longtine MS, Chow CJ, Adames N, Pringle JR, Peter M, Boone C. *Science.* 1997;276:118–122. doi: 10.1126/science.276.5309.118. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
192. Sagot I, Rodal AA, Moseley J, Goode BL, Pellman D. *Nature Cell Biol.* 2002;4:626–631. doi: 10.1038/ncb834. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
193. Matheos D, Metodiev M, Muller E, Stone D, Rose MD. *J Cell Biol.* 2004;165:99–109. doi: 10.1083/jcb.200309089. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
194. Qi M, Elion EA. *J Cell Sci.* 2005;118:2837–48. doi: 10.1242/jcs.02418. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
195. Elion EA, Satterberg B, Kranz JE. *Mol Biol Cell.* 1993;4:495–510. doi: 10.1091/mbc.4.5.495. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
196. Peter M, Gartner A, Horecka J, Ammerer G, Herskowitz I. *Cell.* 1993;73:747–60. doi: 10.1016/0092-8674(93)90254-n. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
197. Tyers M, Fitcher B. *Mol Cell Biol.* 1993;13:5659–69. doi: 10.1128/mcb.13.9.5659. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
198. Peter M, Herskowitz I. *Science.* 1994;265:1228–31. doi: 10.1126/science.8066461. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
199. Gartner A, Jovanovic A, Jeoung DI, Bourlat S, Cross FR, Ammerer G. *Mol Cell Biol.* 1998;18:3681–91. doi: 10.1128/mcb.18.7.3681. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
200. Jeoung DI, Cross F. *Mol. Cells.* 2000;10:460–464. [[PubMed](#)] [[Google Scholar](#)]

201. Escote X, Zapater M, Clotet J, Posas F. *Nature Cell Biol.* 2004;6:997–1002. doi: 10.1038/ncb1174. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
202. Clotet J, Escote X, Adrover MA, Yaakov G, Gari E, Aldea M, de Nadal E, Posas F. *EMBO J.* 2006;25:2338–2346. doi: 10.1038/sj.emboj.7601095. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
203. Alexander MR, Tyers M, Perret M, Craig BM, Fang KS, Gustin MC. *Mol. Biol. Cell.* 2001;12:53–62. doi: 10.1091/mbc.12.1.53. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
204. Zapater M, Clotet J, Escote X, Posas F. *Cell Cycle.* 2005;4:6–7. doi: 10.4161/cc.4.1.1344. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
205. Nash P, Tang X, Orlicky S, Chen Q, Gertler FB, M. M.D., Sicheri F, Pawson T, Tyers M. *Nature.* 2001;414:514–521. doi: 10.1038/35107009. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
206. Kipreos ET, Pagano M. L. Related Articles, and G.B.R.E.N. Review. *Genome Biol.* 2000;1:3002.1–3002.7. [[Google Scholar](#)]
207. Deshaies RJ. *Annu. Rev. Cell Dev. Biol.* 1999;15:435–467. doi: 10.1146/annurev.cellbio.15.1.435. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
208. Willems AR, Goh T, Taylor L, Chernushevich I, Shevchenko A, Tyers M. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 1999;354:1533–1550. doi: 10.1098/rstb.1999.0497. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
209. Barral Y, Parra M, Bidlingmaier, Snyder M. *Genes Dev.* 1999;13:176–187. doi: 10.1101/gad.13.2.176. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
210. McMillan JN, L. M.S., Sia RA, Theesfeld CL, Bardes ES, Pringle JR, Lew DJ. *Mol. Cell. Biol.* 1999;19:6929–6939. doi: 10.1128/mcb.19.10.6929. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
211. Shulewitz MJ, Inouye CJ, Thorner J. *Mol Cell Biol.* 1999;19:7123–37. doi: 10.1128/mcb.19.10.7123. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
212. Harrison JC, Bardes ES, Ohya Y, Lew DJ. *Nat Cell Biol.* 2001;3:417–20. doi:

- 10.1038/35070104. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
213. Errede B, Ammerer G. *Genes Dev.* 1989;3:1349–61. doi: 10.1101/gad.3.9.1349. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
214. Fields A, Herskowitz I. *Cell.* 1985;42:923–930. doi: 10.1016/0092-8674(85)90288-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
215. Roberts CJ, Nelson B, Marton MJ, Stoughton R, Meyer MR, Bennett HA, He YD, Dai H, Walker WL, Hughes TR, Tyers M, Boone C, Friend SH. *Science.* 2000;287:873–80. doi: 10.1126/science.287.5454.873. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
216. Trueheart J, Boeke JD, Fink GR. *Mol. Cell. Biol.* 1987;7:2316–2328. doi: 10.1128/mcb.7.7.2316. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
217. Heiman MG, Walter P J. *Cell Biol.* 2000;151:719–730. doi: 10.1083/jcb.151.3.719. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
218. Dolan JW, Kirkman C, Fields S. *Proc Natl Acad Sci U S A.* 1989;86:5703–7. doi: 10.1073/pnas.86.15.5703. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
219. Van Arsdell SW, Stetler GL, Thorner J. *Mol. Cell. Biol.* 1987;7:749–759. doi: 10.1128/mcb.7.2.749. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
220. Van Arsdell SW, Thorner J. In: *Transcriptional Control Mechanisms*. Granner D, Rosenfeld MG, Chang S, editors. Alan R. Liss, Inc.; New York, NY: 1987. pp. 325–332. [[Google Scholar](#)]
221. Flessel MC, Brake AJ, Thorner J. *Genetics.* 1989;121:223–236. doi: 10.1093/genetics/121.2.223. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
222. Jarvis EE, Hagen DC, Sprague GF, Jr. *Mol. Cell. Biol.* 1988;8:309–320. doi: 10.1128/mcb.8.1.309. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
223. Kronstad JW, Holly JA, MacKay VL. *Cell.* 1987;50:369–377. doi: 10.1016/0092-8674(87)90491-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
224. Yuan YL, Fields S. *Mol. Cell. Biol.* 1991;11:5910–5918. doi: 10.1128/mcb.11.12.5910. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

225. Gajiwala KS, Burley SK. *Curr. Opin. Struct. Biol.* 2000;10:110–116. doi: 10.1016/s0959-440x(99)00057-3. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
226. Pi H, Chien CT, Fields S. *Mol Cell Biol.* 1997;17:6410–8. doi: 10.1128/mcb.17.11.6410. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
227. Lahav R, Gammie A, Tavazoie S, Rose MD. *Mol Cell Biol.* 2007;27:818–829. doi: 10.1128/MCB.00439-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
228. Bickel KS, Morris DR. *Mol Cell Biol.* 2006;26:7901–7912. doi: 10.1128/MCB.01004-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
229. Treisman R, Ammerer G. *Curr Opin Genet Dev.* 1992;2:221–226. doi: 10.1016/s0959-437x(05)80277-1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
230. Carr E, A., Mead J, Vershon AK. *Nucleic Acids Res.* 2004;32:2298–2305. doi: 10.1093/nar/gkh560. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
231. Sengupta P, Cochran BH. *Genes Dev.* 1991;5:1924–34. doi: 10.1101/gad.5.10.1924. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
232. Gammie AE, Stewart BG, Scott CF, Rose MD. *Mol Cell Biol.* 1999;817–825. doi: 10.1128/mcb.19.1.817. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
233. Ren B, Robert F, Wyrick JJ, Aparicio O, Jennings EG, Simon I, Zeitlinger J, Schreiber J, Hannett N, Kanin E, Volkert TL, Wilson CJ, Bell SP, Young RA. *Science.* 2000;290:2306–9. doi: 10.1126/science.290.5500.2306. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
234. Zeitlinger J, Simon I, Harbison CT, Hannett NM, Volkert TL, Fink GR, Young RA. *Cell.* 2003;113:395–404. doi: 10.1016/s0092-8674(03)00301-5. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
235. Hung W, Olson KA, Breitkreutz A, Sadowski I. *Eur J Biochem.* 1997;245:241–51. doi: 10.1111/j.1432-1033.1997.00241.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
236. Song D, Dolan JW, Yuan YL, Fields S. *Genes Dev.* 1991;5:741–50. doi: 10.1101/gad.5.5.741. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
237. Cook JG, Bardwell L, Kron SJ, Thorner J. *Genes Dev.* 1996;10:2831–48. doi: 10.1101/gad.10.22.2831. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

238. Tedford K, Kim S, Sa D, Stevens K, Tyers M. *Curr Biol*. 1997;7:228–38. doi: 10.1016/s0960-9822(06)00118-7. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
239. Chou S, Lane S, Liu H. *Mol Cell Biol*. 2006;26:4794–805. doi: 10.1128/MCB.02053-05. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
240. Gavrias V, Andrianopoulos A, Gimeno CJ, Timberlake WE. *Mol. Microbiol*. 1996;19:1255–1263. doi: 10.1111/j.1365-2958.1996.tb02470.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
241. Madhani HD, Fink GR. *Science*. 1997;275:1314–7. doi: 10.1126/science.275.5304.1314. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
242. Kohler T, Wesche S, Taheri N, Braus GH, Mosch HU. *Eukaryot Cell*. 2002;1:673–86. doi: 10.1128/EC.1.5.673-686.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
243. Lo WS, Dranginis AM. *Mol Biol Cell*. 1998;9:161–71. doi: 10.1091/mbc.9.1.161. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
244. Kim TS, Kim HY, Yoon JH, Kang HS. *Mol. Cell. Biol*. 2004;24:9542–9556. doi: 10.1128/MCB.24.21.9542-9556.2004. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
245. Rupp S, Summers E, Lo HJ, Madhani H, Fink G, Embo J. 1999;18:1257–69. doi: 10.1093/emboj/18.5.1257. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
246. Gagiano M, van Dyk D, Bauer FF, Lambrechts MG, Pretorius IS. *Mol. Microbiol*. 1999;31:103–16. doi: 10.1046/j.1365-2958.1999.01151.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
247. Pan X, Heitman J. *Mol Cell Biol*. 2002;22:3981–93. doi: 10.1128/MCB.22.12.3981-3993.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
248. Kim TS, Lee SB, Kang HS. *Mol Cell Biol*. 2004;24:7695–706. doi: 10.1128/MCB.24.17.7695-7706.2004. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
249. Olson KA, Nelson C, Tai G, Hung W, Yong C, Astell C, Sadowski I. *Mol Cell Biol*.

- 2000;20:4199–209. doi: 10.1128/mcb.20.12.4199-4209.2000. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
250. Bardwell L, Cook JG, Zhu-Shimoni JX, Voora D, Thorner J. Proc Natl Acad Sci U S A. 1998;95:15400–5. doi: 10.1073/pnas.95.26.15400. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
251. Bardwell L, Cook JG, Voora D, Baggott DM, Martinez AR, Thorner J. Genes Dev. 1998;12:2887–98. doi: 10.1101/gad.12.18.2887. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
252. Zakrzewska A, Boorsma A, Brul S, Hellingwerf KJ, Klis FM. Eukaryot Cell. 2005;4:703–715. doi: 10.1128/EC.4.4.703-715.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
253. Watanabe Y, Takaesu G, Hagiwara M, Irie K, Matsumoto K. Mol Cell Biol. 1997;17:2615–23. doi: 10.1128/mcb.17.5.2615. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
254. Ho SW, Jona G, Chen CT, Johnston M, Snyder M. Proc Natl Acad Sci USA. 2006;103:9940–9945. doi: 10.1073/pnas.0509185103. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
255. Yang YL, Suen J, Brynildsen MP, Galbraith SJ, Liao JC. BMC Genomics. 2005;6:90–104. doi: 10.1186/1471-2164-6-90. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
256. Baetz K, Moffat J, Haynes J, Chang M, Andrews B. Mol Cell Biol. 2001;21:6515–28. doi: 10.1128/MCB.21.19.6515-6528.2001. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
257. Rusche LN, Kirchmaier AL, Rine J. Annu Rev Biochem. 2003;481–516. doi: 10.1146/annurev.biochem.72.121801.161547. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
258. Ray A, Hector RE, Roy N, Song JH, Berkner KL, Runge KW. Nat Genet. 2003;33:522–6. doi: 10.1038/ng1132. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
259. Stone EM, Pillus L. J Cell Biol. 1996;135:571–83. doi: 10.1083/jcb.135.3.571.

[[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

260. Rep M, Krantz M, Thevelein JM, Hohmann S. J Biol Chem. 2000;275:8290–300. doi: 10.1074/jbc.275.12.8290. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

261. Posas F, Chambers JR, Heyman JA, Hoeffler JP, de Nadal E, Arino J. J Biol Chem. 2000;275:17249–55. doi: 10.1074/jbc.M910016199. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

262. Rep M, Reiser V, Gartner U, Thevelein JM, Hohmann S, Ammerer G, Ruis H. Mol Cell Biol. 1999;19:5474–85. doi: 10.1128/mcb.19.8.5474. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

263. Edmunds JW, Mahadevan LC. J Cell Sci. 2004;117:3715–23. doi: 10.1242/jcs.01346. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

264. Rep M, Proft M, Remize F, Tamas M, Serrano R, Thevelein JM, Hohmann S. Mol Microbiol. 2001;40:1067–83. doi: 10.1046/j.1365-2958.2001.02384.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

265. Proft M, Serrano R. Mol Cell Biol. 1999;19:537–46. doi: 10.1128/mcb.19.1.537. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

266. Martinez-Pastor MT, Marchler G, Schuller C, Marchler-Bauer A, Ruis H, Estruch F, Embo J. 1996;15:2227–35. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

267. Davie JK, Edmondson DG, Coco CB, Dent SY. J Biol Chem. 2003;278:50158–50162. doi: 10.1074/jbc.M309753200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

268. Green SR, Johnson AD. Mol. Biol. Cell. 2004;15:4191–4202. doi: 10.1091/mbc.E04-05-0412. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

269. Zhang Z, Reese JC. J. Biol. Chem. 2004;279:39240–39250. doi: 10.1074/jbc.M407159200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

270. Proft M, Pascual-Ahuir A, de Nadal E, Arino J, Serrano R, Posas F, Embo J. 2001;20:1123–33. doi: 10.1093/emboj/20.5.1123. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

271. Proft M, Struhl K. *Mol Cell*. 2002;9:1307–17. doi: 10.1016/s1097-2765(02)00557-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
272. Alepuz PM, Jovanovic A, Reiser V, Ammerer G. *Mol Cell*. 2001;7:767–77. doi: 10.1016/s1097-2765(01)00221-0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
273. Alepuz PM, de Nadal E, Zapater M, Ammerer G, Posas F, Embo J. 2003;22:2433–42. doi: 10.1093/emboj/cdg243. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
274. De Nadal E, Zapater M, Alepuz PM, Sumoy L, Mas G, Posas F. *Nature*. 2004;427:370–4. doi: 10.1038/nature02258. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
275. Pokholok DK, Zeitlinger J, Hannett NM, Reynolds DB, Young RA. *Science*. 2006;313:533–6. doi: 10.1126/science.1127677. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
276. Proft M, Mas G, de Nadal E, Vendrell A, Noriega N, Struhl K, Posas F. *Mol Cell*. 2006;23:241–50. doi: 10.1016/j.molcel.2006.05.031. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
277. Chow CW, Davis RJ. *Cell*. 2006;127:887–890. doi: 10.1016/j.cell.2006.11.015. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
278. Varela JC, van Beekvelt C, Planta RJ, Mager WH. *Mol Microbiol*. 1992;6:2183–90. doi: 10.1111/j.1365-2958.1992.tb01392.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
279. Uesono Y, Toh EA. *J Biol Chem*. 2002;277:13848–55. doi: 10.1074/jbc.M108848200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
280. Jiang L, Niu S, Clines KL, Burke DJ, Sturgill TW. *Mol Genet Genomics*. 2004;271:208–19. doi: 10.1007/s00438-003-0972-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
281. Melcher ML, Thorner J. *J Biol Chem*. 1996;271:29958–68. doi: 10.1074/jbc.271.47.29958. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
282. Nairn AC, Palfrey HC. *J Biol Chem*. 1987;262:17299–303. [[PubMed](#)] [[Google Scholar](#)]
283. Law GL, Bickel KS, MacKay VL, Morris DR. *Genome Biol*. 2005;6:R111. doi: 10.1186/gb-2005-6-13-r111. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

284. MacKay VL, Li X, Flory MR, Turcott E, Law GL, Serikawa KA, Xu XL, Lee H, Goodlett DR, Aebersold R, Zhao LP. *Mol Cell Proteomics*. 2004;3:478–489. doi: 10.1074/mcp.M300129-MCP200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
285. Serikawa KA, Xu XL, MacKay CL, L. G.L., Zong Q, Zhao LP, Bumgarner R, Morris DR. *Mol Cell Proteomics*. 2003;2:191–204. doi: 10.1074/mcp.D200002-MCP200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
286. Ashe MP, De Long SK, Sachs AB. *Mol Biol Cell*. 2000;11:833–848. doi: 10.1091/mbc.11.3.833. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
287. Vasudevan S, Peltz SW. *Mol Cell*. 2001;7:1191–200. doi: 10.1016/s1097-2765(01)00279-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
288. Vasudevan S, Garneau N, Tu Khounh D, Peltz SW. *Mol Cell Biol*. 2005;25:9753–63. doi: 10.1128/MCB.25.22.9753-9763.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
289. Duttagupta R, Tian B, Wilusz CJ, Khounh DT, S. P, Ouyang M, Dougherty JP, Peltz SW. *Mol Cell Biol*. 2005;25:5499–5513. doi: 10.1128/MCB.25.13.5499-5513.2005. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
290. Sachs AB, Sarnow P, Hentze MW. *Cell*. 1997;89:831–838. doi: 10.1016/s0092-8674(00)80268-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
291. Park YU, Hur H, Ka M, Kim J. *Eukaryot Cell*. 2006 doi: 10.1128/EC.00121-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
292. Brengues M, Teixeira D, Parker R. *Science*. 2005;310:486–489. doi: 10.1126/science.1115791. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
293. Ibrahimo S, H. L.E., Ashe MP. *Yeast*. 2006;23:1075–1088. doi: 10.1002/yea.1415. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
294. Esch RK, Wang Y, Errede B. *Eukaryot Cell*. 2006 doi: 10.1128/EC.00270-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
295. Nelson C, Goto S, Lund K, Hung W, Sadowski I. *Nature*. 2003;421:187–90. doi: 10.1038/nature01243. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

296. Sheu YJ, Santos B, Fortin N, Costigan C, Snyder M. *Mol Cell Biol.* 1998;18:4053–69. doi: 10.1128/mcb.18.7.4053. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
297. van Drogen F, Peter M. *Curr Biol.* 2002;12:1698–703. doi: 10.1016/s0960-9822(02)01186-7. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
298. Elion EA. *J Cell Sci.* 2001;114:3967–78. doi: 10.1242/jcs.114.22.3967. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
299. Dard N, Peter M. *Bioessays.* 2006;28:146–156. doi: 10.1002/bies.20351. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
300. Mahanty SK, Wang Y, Farley FW, Elion EA. *Cell.* 1999;98:501–12. doi: 10.1016/s0092-8674(00)81978-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
301. Pryciak PM, Huntress FA. *Genes Dev.* 1998;12:2684–97. doi: 10.1101/gad.12.17.2684. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
302. van Drogen F, Stucke VM, Jorritsma G, Peter M. *Nat Cell Biol.* 2001;3:1051–9. doi: 10.1038/ncb1201-1051. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
303. Sette C, Inouye CJ, Stroschein SL, Iaquina PJ, Thorner J. *Mol Biol Cell.* 2000;11:4033–49. doi: 10.1091/mbc.11.11.4033. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
304. Lamson RE, Takahashi S, Winters MJ, Pryciak PM. *Curr Biol.* 2006;16:618–23. doi: 10.1016/j.cub.2006.02.060. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
305. Wang Y, Chen W, Simpson DM, Elion EA. *J Biol Chem.* 2005;280:13084–96. doi: 10.1074/jbc.M410461200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
306. Shimada Y, Wiget P, Gulli MP, Bi E, Peter M. *EMBO J.* 2004;23:1051–1062. doi: 10.1038/sj.emboj.7600124. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
307. Inouye C, Dhillon N, Durfee T, Zambryski PC, Thorner J. *Genetics.* 1997;147:479–92. doi: 10.1093/genetics/147.2.479. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
308. Yablonski D, Marbach I, Levitzki A. *Proc Natl Acad Sci U S A.* 1996;93:13864–9. doi:

- 10.1073/pnas.93.24.13864. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
309. Wang Y, Elion EA. *Mol Biol Cell*. 2003;14:2543–58. doi: 10.1091/mbc.E02-10-0699. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
310. Bhattacharyya RP, Remenyi A, Good MC, Bashor CJ, Falick AM, Lim WA. *Science*. 2006;311:822–6. doi: 10.1126/science.1120941. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
311. Gartner A, Nasmyth K, Ammerer G. *Genes Dev*. 1992;6:1280–92. doi: 10.1101/gad.6.7.1280. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
312. Kranz JE, Satterberg B, Elion EA. *Genes Dev*. 1994;8:313–27. doi: 10.1101/gad.8.3.313. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
313. Flotho A, Simpson DM, Qi M, Elion EA. *J Biol Chem*. 2004;279:47391–401. doi: 10.1074/jbc.M405681200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
314. van Drogen F, Peter M. *Biol Cell*. 2001;93:63–70. doi: 10.1016/s0248-4900(01)01123-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
315. Burack WR, Shaw AS. *J Biol Chem*. 2005;280:3832–3837. doi: 10.1074/jbc.M410031200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
316. Chen RH, Sarnecki C, Blenis J. *Mol Cell Biol*. 1992;12:915–927. doi: 10.1128/mcb.12.3.915. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
317. Khokhlatchev AV, Canagarajah B, Wilsbacher J, Robinson M, Atkinson M, Goldsmith E, Cobb MH. *Cell*. 1998;93:605–615. doi: 10.1016/s0092-8674(00)81189-7. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
318. Adachi M, Fukuda M, Nishida E. *Embo J*. 1999;18:5347–58. doi: 10.1093/emboj/18.19.5347. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
319. Matsubayashi Y, Fukuda M, Nishida E. *J Biol Chem*. 2001;276:41755–60. doi: 10.1074/jbc.M106012200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
320. Whitehurst AW, Wilsbacher JL, You Y, Luby-Phelps K, Moore MS, Cobb MH. *Proc Natl Acad Sci U S A*. 2002;99:7496–501. doi: 10.1073/pnas.112495999. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

321. Kondoh K, Torii S, Nishida E. *Chromosoma*. 2005;114:86–91. doi: 10.1007/s00412-005-0341-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
322. Formstecher E, Ramos JW, Fauquet M, Calderwood DA, Hsieh JC, Canton B, Nguyen XT, Barnier JV, Camonis J, Ginsberg MH, Chneiweiss H. *Dev Cell*. 2001;1:239–50. doi: 10.1016/s1534-5807(01)00035-1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
323. Whitehurst AW, Robinson FL, Moore MS, Cobb MH. *J Biol Chem*. 2004;279:12840–12847. doi: 10.1074/jbc.M310031200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
324. Reiser V, Ruis H, Ammerer G. *Mol Biol Cell*. 1999;10:1147–61. doi: 10.1091/mbc.10.4.1147. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
325. Westfall PJ, Thorner J. *Eukaryot Cell*. 2006;5:1215–28. doi: 10.1128/EC.00037-06. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
326. Mattison CP, Ota IM. *Genes Dev*. 2000;14:1229–35. [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
327. Blackwell E, Halatek IM, Kim HJ, Ellicott AT, Obukhov AA, Stone DE. *Mol Cell Biol*. 2003;23:1135–50. doi: 10.1128/MCB.23.4.1135-1150.2003. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
328. Metodiev MV, Matheos D, Rose MD, Stone DE. *Science*. 2002;296:1483–6. doi: 10.1126/science.1070540. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
329. Marshall CJ. *Cell*. 1995;80:179–85. doi: 10.1016/0092-8674(95)90401-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
330. Bhalla US, Ram PT, Iyengar R. *Science*. 2002;297:1018–23. doi: 10.1126/science.1068873. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
331. Wang X, Hao N, Dohlman HG, Elston TC. *Biophys. J*. 2006;90:1961–1978. doi: 10.1529/biophysj.105.073874. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
332. Keyse SM. *Curr. Opin. Cell Biol*. 2000;12:186–192. doi: 10.1016/s0955-0674(99)00075-7. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
333. Martin H, Flandez M, Nombela C, Molina M. *Mol Microbiol*. 2005;58:6–16. doi: 10.1111/j.1365-2958.2005.04822.x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

334. Doi K, Gartner A, Ammerer G, Errede B, Shinkawa H, Sugimoto K, Matsumoto K. *Embo J*. 1994;13:61–70. doi: 10.1002/j.1460-2075.1994.tb06235.x. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
335. Maeda T, Wurgler-Murphy SM, Saito H. *Nature*. 1994;369:242–5. doi: 10.1038/369242a0. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
336. Jacoby T, Flanagan H, Faykin A, Seto AG, Mattison C, Ota I. *J Biol Chem*. 1997;272:17749–55. doi: 10.1074/jbc.272.28.17749. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
337. Wurgler-Murphy SM, Maeda T, Witten EA, Saito H. *Mol Cell Biol*. 1997;17:1289–97. doi: 10.1128/mcb.17.3.1289. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
338. Zhan XL, Deschenes RJ, Guan KL. *Genes Dev*. 1997;11:1690–702. doi: 10.1101/gad.11.13.1690. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
339. Mattison CP, Spencer SS, Kresge KA, Lee J, Ota IM. *Mol Cell Biol*. 1999;19:7651–60. doi: 10.1128/mcb.19.11.7651. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
340. Warmka J, Hanneman J, Lee J, Amin D, Ota I. *Mol Cell Biol*. 2001;21:51–60. doi: 10.1128/MCB.21.1.51-60.2001. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
341. Collister M, Didmon MP, MacIsaac F, Stark MJ, MacDonald NQ, Keyse SM. *FEBS Lett*. 2002;527:186–92. doi: 10.1016/s0014-5793(02)03220-9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
342. Young C, Mapes J, Hanneman J, Al-Zarban S, Ota I. *Eukaryot Cell*. 2002;1:1032–40. doi: 10.1128/EC.1.6.1032-1040.2002. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
343. Hahn JS, Thiele DJ. *J Biol Chem*. 2002;277:21278–84. doi: 10.1074/jbc.M202557200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
344. Flandez M, Cosano IC, Nombela C, Martin H, Molina M. *J Biol Chem*. 2004;279:11027–34. doi: 10.1074/jbc.M306412200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
345. Mapes J, Ota IM. *Embo J*. 2004;23:302–11. doi: 10.1038/sj.emboj.7600036. [[DOI](#)]

[\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

346. Sato N, Kawahara H, Toh-e A, Maeda T. *Mol Cell Biol*. 2003;23:6662–71. doi: 10.1128/MCB.23.18.6662-6671.2003. [\[DOI\]](#)] [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

347. Wang Y, Dohlman HG. *J Biol Chem*. 2002;277:15766–72. doi: 10.1074/jbc.M111733200. [\[DOI\]](#)] [\[PubMed\]](#) [\[Google Scholar\]](#)]

348. Wang Y, Ge Q, Houston D, Thorner J, Errede B, Dohlman HG. *J Biol Chem*. 2003;278:22284–9. doi: 10.1074/jbc.M301272200. [\[DOI\]](#)] [\[PubMed\]](#) [\[Google Scholar\]](#)]

349. Schaubert C, Chen L, Tongaonkar P, Vega I, Madura K. *Genes Cells*. 1998;3:307–319. doi: 10.1046/j.1365-2443.1998.00192.x. [\[DOI\]](#)] [\[PubMed\]](#) [\[Google Scholar\]](#)]

350. Wang Y, Marotti LA, Jr., Lee MJ, Dohlman HG. *J Biol Chem*. 2005;280:284–91. doi: 10.1074/jbc.M411624200. [\[DOI\]](#)] [\[PubMed\]](#) [\[Google Scholar\]](#)]

351. Esch RK, Errede B. *Proc Natl Acad Sci U S A*. 2002;99:9160–5. doi: 10.1073/pnas.142034399. [\[DOI\]](#)] [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

352. Dohlman HG, Song J, Ma D, Courchesne WE, Thorner J. *Mol Cell Biol*. 1996;16:5194–209. doi: 10.1128/mcb.16.9.5194. [\[DOI\]](#)] [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

353. Chasse SA, Flanary P, Parnell SC, Hao N, Cha JY, Siderovski DP, Dohlman HG. *Eukaryot Cell*. 2006;5:330–46. doi: 10.1128/EC.5.2.330-346.2006. [\[DOI\]](#)] [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

354. Ballon DR, Flanary PL, Gladue DP, Konopka JB, Dohlman HG, Thorner J. *Cell*. 2006;126:1079–93. doi: 10.1016/j.cell.2006.07.030. [\[DOI\]](#)] [\[PubMed\]](#) [\[Google Scholar\]](#)]

355. Chen T, Kurjan J. *Mol Cell Biol*. 1997;17:3429–39. doi: 10.1128/mcb.17.6.3429. [\[DOI\]](#)] [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

356. Xu BE, Skowronek KR, Kurjan J. *Genetics*. 2001;159:1559–71. doi: 10.1093/genetics/159.4.1559. [\[DOI\]](#)] [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)]

357. Burchett SA, Flanary P, Aston C, Jiang L, Young KH, Uetz P, Fields S, Dohlman HG. *J Biol Chem*. 2002;277:22156–67. doi: 10.1074/jbc.M202254200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
358. Hicke L, Dunn R. *Annu Rev Cell Dev Biol*. 2003;19:141–172. doi: 10.1146/annurev.cellbio.19.110701.154617. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
359. Strickfaden SC, Winters MJ, Ben-Ari G, Lamson RE, Tyers M, Pryciak PM. *Cell*. 2007;128:519–531. doi: 10.1016/j.cell.2006.12.032. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
360. Serber Z, Ferrell JEJ. *Cell*. 2007;128:441–444. doi: 10.1016/j.cell.2007.01.018. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
361. Kunzler M, Trueheart J, Sette C, Hurt E, Thorner J. *Genetics*. 2001;157:1089–105. doi: 10.1093/genetics/157.3.1089. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
362. Frey S, Pool M, Seedorf M. *J Biol Chem*. 2001;276:15905–15912. doi: 10.1074/jbc.M009430200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
363. Lang BD, Fridovich-Keil JL. *Nucleic Acids Res*. 2000;28:1576–1584. doi: 10.1093/nar/28.7.1576. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
364. Hartwell LH. *J Cell Biol*. 1980;85:811–822. doi: 10.1083/jcb.85.3.811. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
365. Dietzel C, Kurjan J. *Cell*. 1987;50:1001–1010. doi: 10.1016/0092-8674(87)90166-8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
366. Miyajima I, Nakafuku M, Nakayama N, Brenner C, Miyajima A, Kaibuchi K, Arai K, Kaziro Y, Matsumoto K. *Cell*. 1987;50:1011–1019. doi: 10.1016/0092-8674(87)90167-x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
367. Miyajima I, Nakayama N, Nakafuku M, Kaziro Y, Arai K, Matsumoto K. *Genetics*. 1988;119:797–804. doi: 10.1093/genetics/119.4.797. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
368. Breitkreutz A, Tyers M. *Trends Cell Biol*. 2002;12:254–7. doi: 10.1016/s0962-

- 8924(02)02284-5. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
369. Schwartz MA, Madhani HD. *Annu Rev Genet.* 2004;38:725–48. doi: 10.1146/annurev.genet.39.073003.112634. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
370. Elion EA, Qi M, Chen W. *Science.* 2005;307:687–8. doi: 10.1126/science.1109500. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
371. Bardwell L. *Biochem Soc Trans.* 2006;34:837–41. doi: 10.1042/BST0340837. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
372. O'Rourke SM, Herskowitz I. *Genes Dev.* 1998;12:2874–86. doi: 10.1101/gad.12.18.2874. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
373. Enslen H, Davis RJ. *Biol Cell.* 2001;93:5–14. doi: 10.1016/s0248-4900(01)01156-x. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
374. Tanoue T, Nishida E. *Cell Signal.* 2003;15:455–62. doi: 10.1016/s0898-6568(02)00112-2. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
375. Marles JA, Dahesh S, Haynes J, Andrews BJ, Davidson AR. *Mol Cell.* 2004;14:813–23. doi: 10.1016/j.molcel.2004.05.024. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
376. Zarrinpar A, Park SH, Lim WA. *Nature.* 2003;426:676–80. doi: 10.1038/nature02178. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
377. Seet BT, Pawson T. *Curr Biol.* 2004;14:R708–10. doi: 10.1016/j.cub.2004.08.044. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
378. Harris K, Lamson RE, Nelson B, Hughes TR, Marton MJ, Roberts CJ, Boone C, Pryciak PM. *Curr Biol.* 2001;11:1815–24. [[PubMed](#)] [[Google Scholar](#)]
379. Harris K, Lamson RE, Nelson B, Hughes TR, Marton MJ, Roberts CJ, Boone C, Pryciak PM. *Curr. Biol.* 2001;11:1815–1824. [[PubMed](#)] [[Google Scholar](#)]
380. Park SH, Zarrinpar A, Lim WA. *Science.* 2003;299:1061–4. doi: 10.1126/science.1076979. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
381. Bardwell L, Cook JG, Chang EC, Cairns BR, Thorner J. *Mol Cell Biol.* 1996;16:3637–50. doi: 10.1128/mcb.16.7.3637. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google](#)]

[Scholar](#)]

382. Bardwell L, Thorner J. Trends Biochem Sci. 1996;21:373–4. [[PubMed](#)] [[Google Scholar](#)]

383. Schwartz MA, Madhani HD. Curr Genet. 2006;49:351–63. doi: 10.1007/s00294-006-0061-6. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

384. Jansen G, Buhning F, Hollenberg CP, Ramezani Rad M. Mol Genet Genomics. 2001;265:102–17. doi: 10.1007/s004380000394. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

385. Wu C, Arcand M, Jansen G, Zhong M, Iouk T, Thomas DY, Meloche S, Whiteway M. Eukaryot Cell. 2003;2:949–61. doi: 10.1128/EC.2.5.949-961.2003. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

386. Bruckner S, Kohler T, Braus GH, Heise B, Bolte M, Mosch HU. Curr Genet. 2004;46:331–42. doi: 10.1007/s00294-004-0545-1. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

387. Wang Y, Dohlman HG. J Biol Chem. 2006;281:1964–9. doi: 10.1074/jbc.M508985200. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

388. Bossis G, Melchior F. Cell Div. 2006;1:13–20. doi: 10.1186/1747-1028-1-13. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

389. Buehrer BM, Errede B. Mol Cell Biol. 1997;17:6517–25. doi: 10.1128/mcb.17.11.6517. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]